Designing a RISC-V SoC

Course material for subject Digital Design

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Preface

This material was written alongside the lectures for the course DN, redesigned around the theme of RISC-V SoC design. Each chapter corresponds to one week of lectures and serves a dual purpose:

1. As **lecture notes** — rendered as a slide deck (reveal.js).
2. As a **book chapter** — rendered as part of the combined PDF/HTML course material.

Content intended for only one of the two formats (e.g. discussion questions used in class, or in-depth derivations meant for the book) is marked with `content-visible` blocks — see the example in the chapter on pipeline structure.

> ⚠ Work in progress
>
> This material is a work in progress. It is written and revised alongside the course as it is taught, so chapters vary in polish, and some sections still need further revision. Please read it with that in mind, and feel free to flag anything that is unclear, incomplete, or incorrect.

1 Introduction to Digital Design

Introduction to Digital Design

1.1 Goal for this week

- See why the P-N junction's one-way conduction is the basic building block behind every diode and every transistor
- Follow how a MOSFET's gate voltage creates, and destroys, a conducting channel between source and drain, turning it into a voltage-controlled switch
- Build the CMOS inverter from one NMOS and one PMOS, and see why that pairing is what gives CMOS its low static power
- Separate static and dynamic power, and see which design choices trade one for the other
- Connect the transistor-level picture to the chip-level one: how NMOS and PMOS actually end up side by side on the same piece of silicon, and roughly how a wafer becomes a chip

1.2 The P-N junction

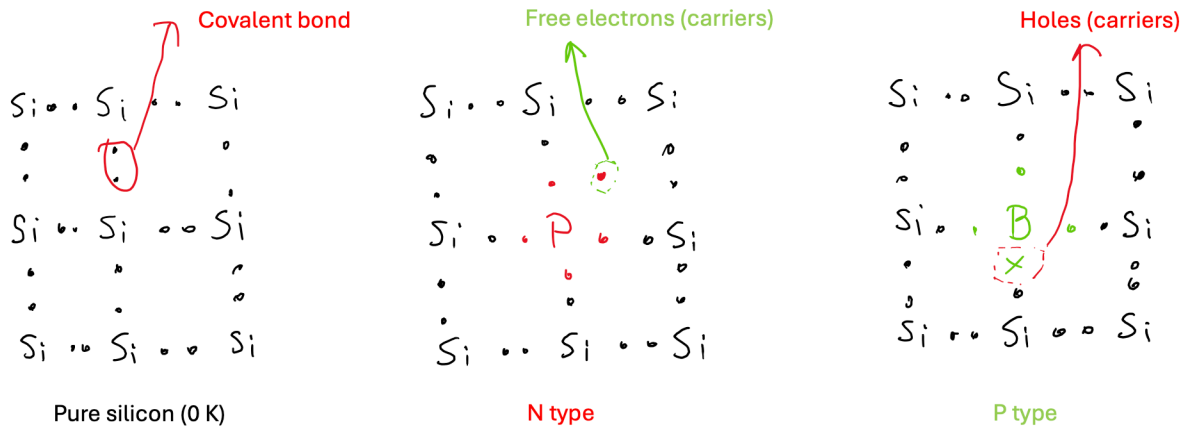


Figure 1.1: N-type and P-type silicon: doping replaces some silicon atoms with impurities that donate either a free electron or a hole

One of the fundamental facts of solid-state electronics is how a P-N junction behaves. On its own, pure silicon is a poor conductor. We fix that through doping, deliberately introducing a small number of impurity atoms into the silicon lattice to change how many free charge carriers it has. Dope silicon with an element that has more valence electrons than silicon, such as phosphorus or arsenic, and the extra electrons become free carriers. We call the result n-type silicon, “n” for the negative charge of its majority carrier. Dope it instead with an element that has fewer valence electrons, such as boron or gallium, and the missing electron leaves behind a “hole”, a position that behaves like a mobile positive charge. We call that p-type silicon.

Bring a block of n-type and a block of p-type silicon into contact, and the junction between them behaves in a way neither material does alone. Current flows easily in one direction, forward bias, and is almost completely blocked in the other, reverse bias. This one-way behavior is not a separate property we add. It falls directly out of how charge moves across the junction, and it is the basic mechanism behind every diode, and in turn, every transistor we build in this course.

1.2.1 The depletion region

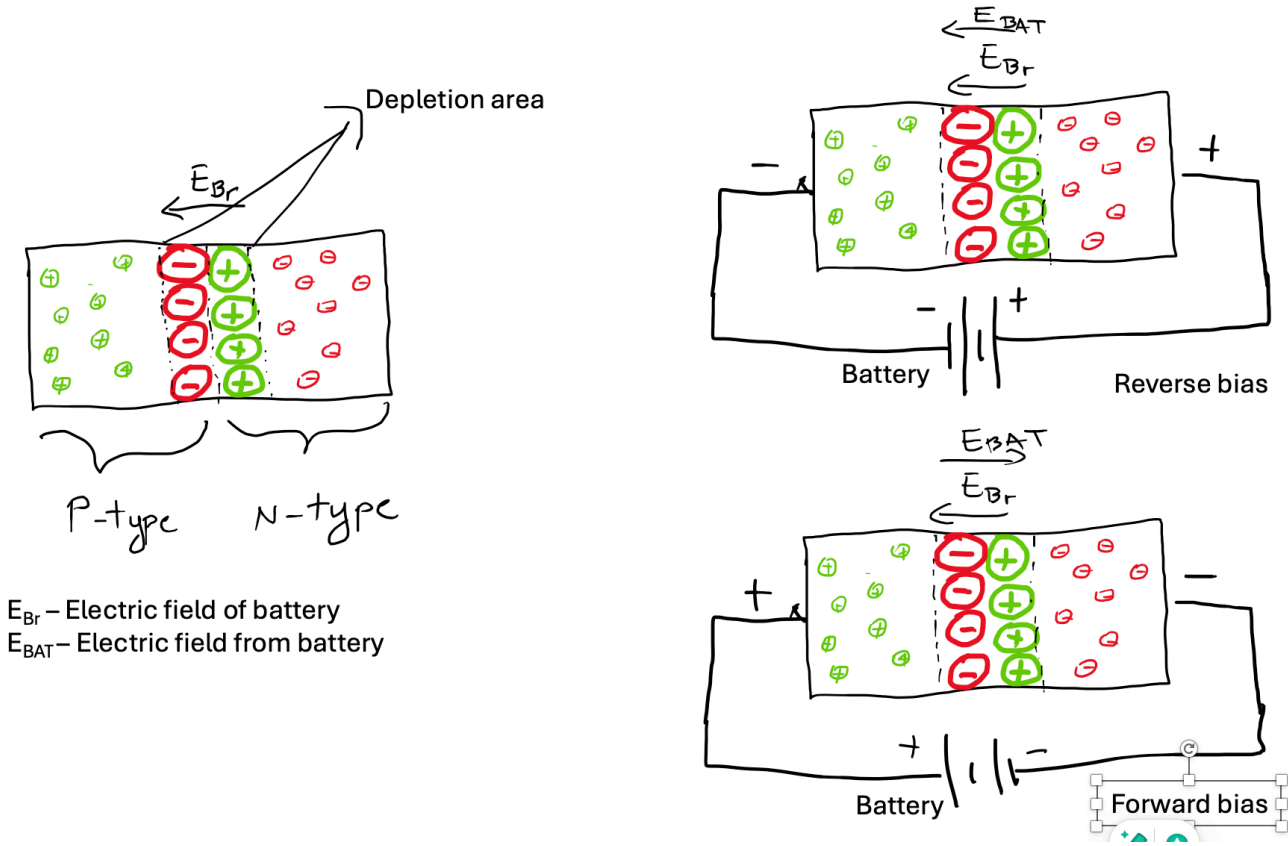


Figure 1.2: The depletion region at a P-N junction, and how reverse bias and forward bias each move the electric field relative to the battery's field

Right at the junction, free electrons from the n-side and free holes from the p-side meet and recombine, leaving behind a thin region with no free carriers at all. We call this the depletion region. What is left behind there is not neutral. It is the fixed, immobile dopant ions that donated those carriers in the first place, and those fixed charges set up an electric field across the depletion region, pointing from the n-side toward the p-side.

1.2.1.1 Equilibrium

With no external voltage applied, that built-in field reaches a balance point. It is strong enough to stop any further electrons or holes from diffusing across the junction, but no stronger. We call this the equilibrium state. Electrically, it behaves like a small built-in

barrier, often called the Coulomb barrier, that a carrier has to overcome before it can cross from one side to the other.

1.2.1.2 Reverse bias

Apply an external voltage with its positive terminal on the n-side, and that external field points the same way as the junction's built-in field. The two add together, the depletion region widens, and the barrier an electron would need to cross gets taller, not shorter. For current to flow at all, electrons from the n-side would have to reach the junction and recombine with holes on the p-side, and a reverse-biased junction actively drives electrons away from the junction instead. The result is that almost no current flows.

1.2.1.3 Forward bias

Flip the polarity, and the external field now opposes the built-in field instead of reinforcing it. The depletion region shrinks, the Coulomb barrier collapses, and carriers can cross the junction with very little resistance. This is forward bias, and it is the conducting state of a diode.

1.3 CMOS technology

Complementary Metal-Oxide-Semiconductor (CMOS) technology is, practically speaking, unavoidable in digital design. It builds every logic function from paired n-type and p-type MOSFETs, and that pairing is what lets it combine very low static power with high noise immunity and fast switching, all at once. That combination is exactly what large-scale integration needs, whether that means a battery-powered microcontroller or the millions of logic gates on a modern microprocessor. Every digital building block we design in this course, down to the CLB we met in the FPGA fabric, ultimately bottoms out in CMOS transistors like the ones in this chapter.

1.3.1 MOSFET transistors

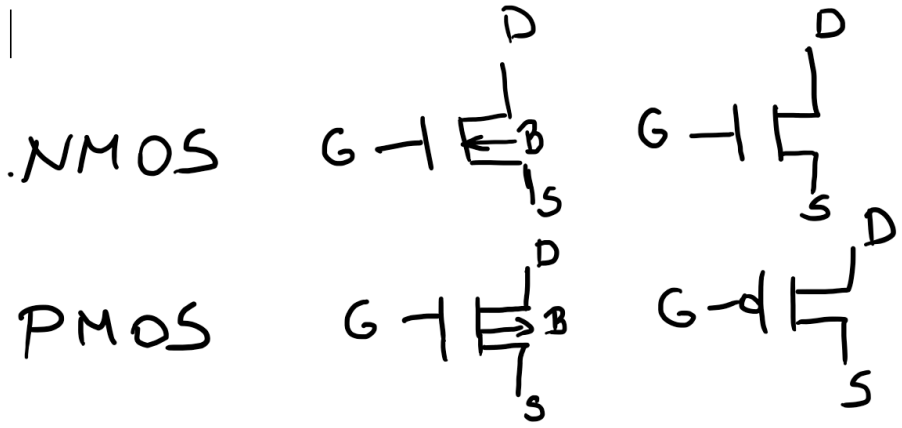


Figure 1.3: NMOS and PMOS schematic symbols: the four-terminal version shows the body connection explicitly, the three-terminal version assumes it

We will mostly use the MOSFET as a switch, though it can also amplify signals. At first glance it looks like a four-terminal device: gate, source, drain, and body. A voltage on the gate decides whether, and how strongly, current can flow between source and drain. The body terminal shifts the device's voltage-current characteristics and is normally just tied to a fixed supply rail, which is why many schematic symbols leave it out and show only three terminals.

The name spells out what the device is built from. "Metal-Oxide- Semiconductor" describes its physical structure. "Field-Effect Transistor" describes how it works: an electric field, not a direct physical connection, controls the current.

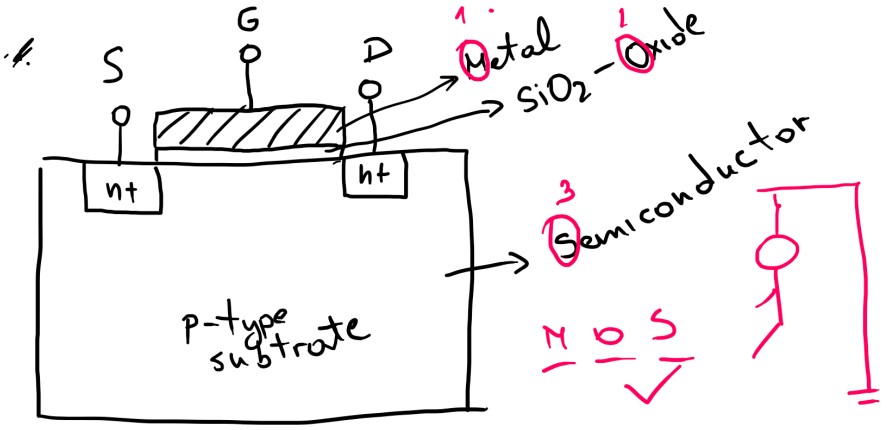


Figure 1.4: NMOS cross-section: a metal gate over an oxide insulator, sitting above a p-type substrate with two n+ regions for source and drain

We illustrate an NMOS. Its gate is a metal layer separated from the p-type substrate beneath it by a thin oxide layer, which insulates the gate and prevents any DC current from flowing into it. Source and drain are both n+ regions embedded directly in that p-type substrate. The names describe their role in conduction: the source supplies the charge carriers, electrons for an n-channel device and holes for a p-channel one, that flow through the channel, and the drain is where those carriers leave it.

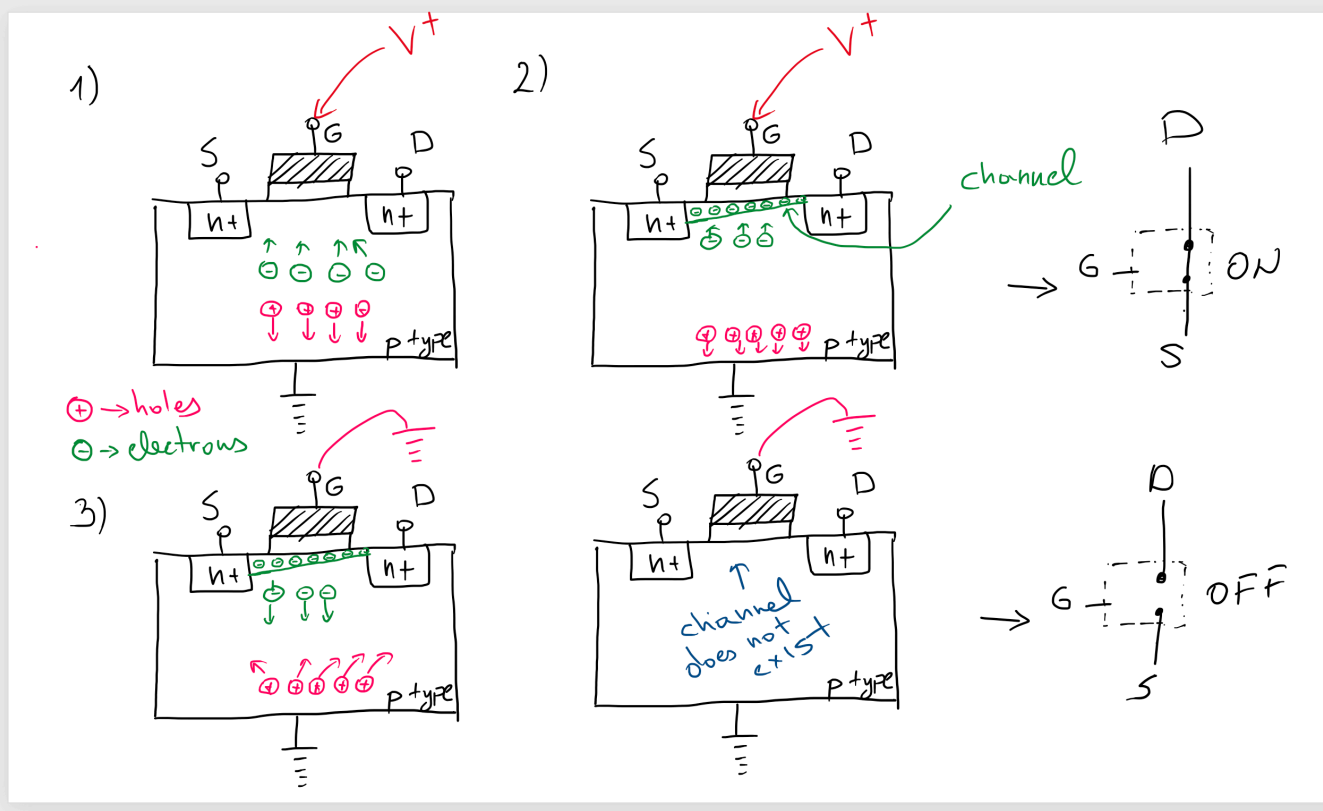


Figure 1.5: Turning an NMOS on: a positive gate voltage pulls electrons to the surface and forms a conducting channel between source and drain, which collapses again once the gate voltage is removed

To turn an NMOS on, we apply a positive voltage to the gate. That voltage attracts electrons from the p-type substrate toward the surface and pushes holes downward and away, purely through the electric field it creates, which is exactly what makes this a field-effect transistor. Once enough electrons have gathered at the surface, they form a bridge, the channel, connecting source and drain, and current can flow between them. How long that channel is, geometrically, is a property of the fabrication technology used.

In effect, the MOS transistor behaves as a controlled switch. Take the voltage off the gate, and the electrons no longer have anywhere to stay. The channel dissolves, and the connection between drain and source opens back up.

1.3.2 Complementary MOS

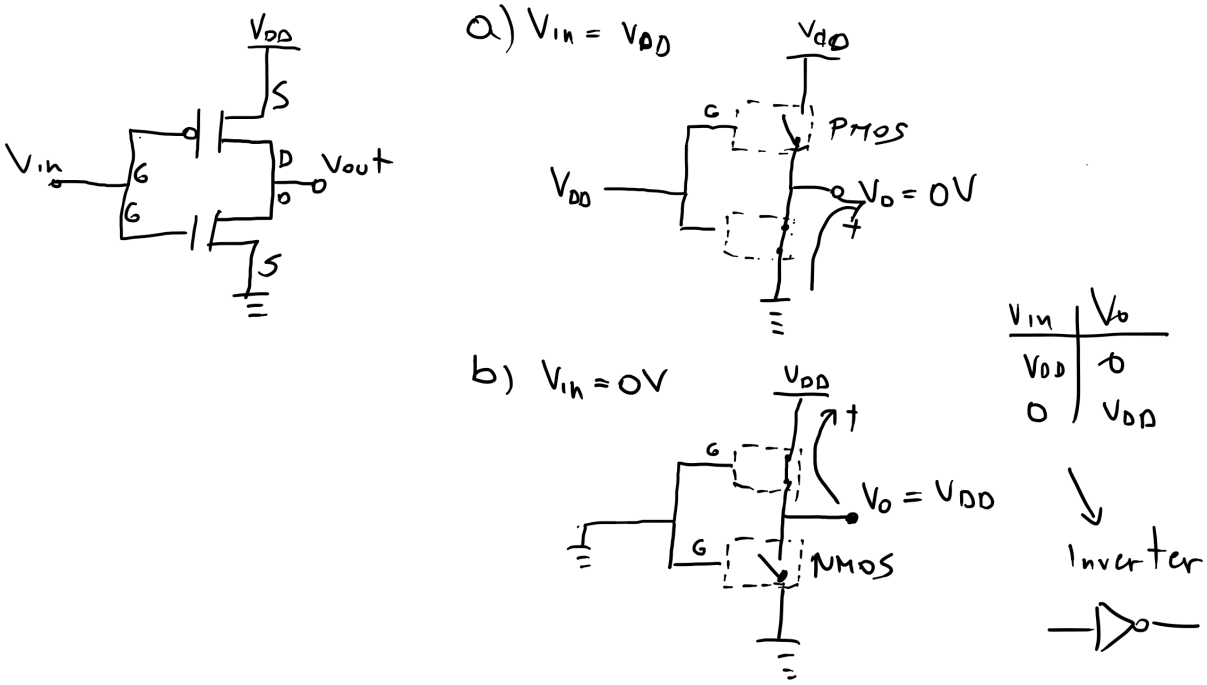


Figure 1.6: CMOS inverter: tying NMOS and PMOS gates together means exactly one of them conducts for any input, pulling the output to either ground or VDD

A CMOS gate pairs an NMOS and a PMOS transistor that share a common gate input. We already know how to turn an NMOS on, and a PMOS is its mirror image: it turns on when the gate is low rather than high. Tie a positive voltage to the shared gate, and the NMOS turns on while the PMOS turns off, pulling the output down to ground. Tie the gate to ground instead, and the PMOS turns on while the NMOS turns off, pulling the output up to VDD. In both cases, exactly one of the two transistors conducts, never both, and the output ends up at the opposite logic level from the input. That is an inverter, and it is the simplest possible CMOS gate.

1.4 CMOS characteristics

1.4.1 Transfer function

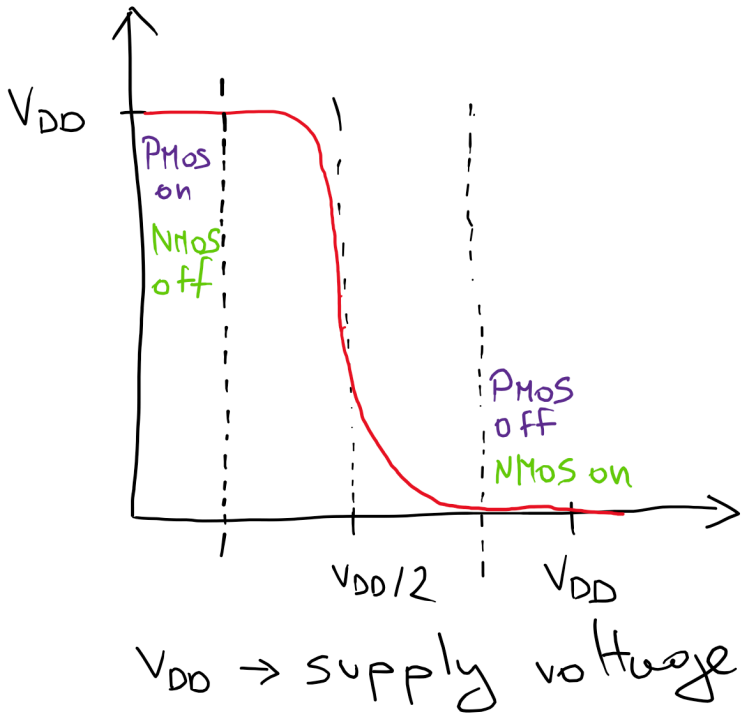


Figure 1.7: CMOS inverter transfer function: the output stays near VDD or near ground across most of the input range, and switches abruptly near VDD/2

The transfer function shows how the output voltage depends on the input voltage. For a CMOS inverter, it is almost a step function. For a wide range of low inputs, the PMOS is on and the NMOS is off, and the output sits close to VDD. For a wide range of high inputs, the roles reverse and the output sits close to ground. Only in a narrow band around VDD/2 do both transistors conduct at once, and the output swings sharply from one rail to the other. That sharp transition is exactly what we want from a digital gate. It is what keeps a “1” looking like a clean 1 and a “0” looking like a clean 0, right up until the input crosses the switching threshold.

1.4.2 Noise margins

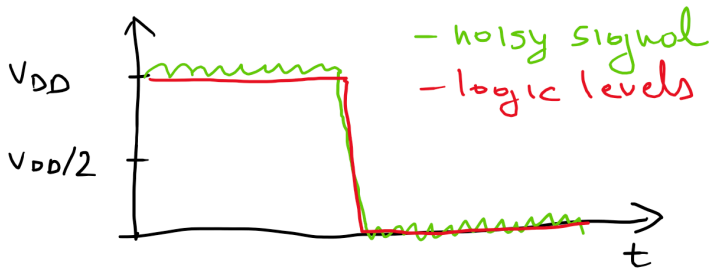


Figure 1.8: A noisy signal riding on a CMOS logic level. As long as the noise stays within the flat regions of the transfer function, it never gets interpreted as the wrong logic level

Because the transfer function stays flat near VDD and near ground across most of the input range, a CMOS gate tolerates a surprising amount of noise. A signal can pick up noise from neighboring wires or supply fluctuations and still be read correctly, as long as that noise does not push the voltage past the switching threshold in the middle of the transfer function. This margin between “close enough to a clean logic level” and “close enough to flip the gate’s decision” is the noise margin, and CMOS’s wide noise margins are one of the main reasons it dominates digital design.

1.4.3 Power

CMOS technology is known for its low power consumption, which is one of its biggest advantages over older logic families. We split that power consumption into two categories: static power and dynamic power.

Static power (also called leakage power) is what the circuit consumes while it is not switching at all. It comes mostly from small leakage currents through transistors that are nominally off, and it becomes a much bigger concern as transistor geometries shrink.

Dynamic power is what the circuit consumes while it is actively switching. Every time a gate’s output toggles, we have to charge or discharge its load capacitance, and that costs energy. We can express the dynamic power of a single node as

$$P_{dynamic} = \alpha C V^2 f$$

where α is the activity factor (the fraction of clock cycles in which the node actually switches), C is the load capacitance, V is the supply voltage, and f is the switching frequency.

1.4.4 Dynamic behavior

A CMOS gate does not switch instantaneously. To change its output, it first has to charge the load capacitance, when the output rises, or discharge it, when the output falls. That load capacitance is not one single thing. It is the sum of the input capacitance of every gate the output drives, the capacitance of the interconnect wiring between them, and the capacitance of the reverse-biased P-N junctions inside the transistors themselves. To make a gate switch faster, we have a few levers: raise the supply voltage, widen the transistor's channel width-to-length ratio, or simply drive fewer downstream gates from the same output.

1.5 Manufacturing CMOS integrated circuits

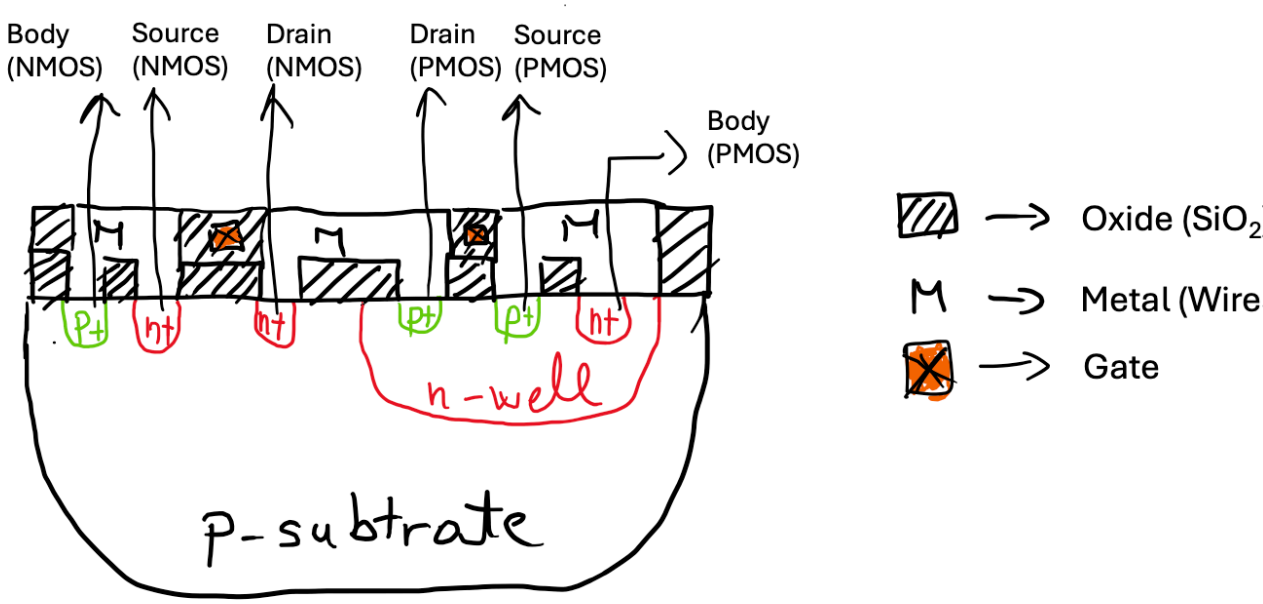


Figure 1.9: Cross-section of a CMOS chip: the PMOS sits inside an n-well so it can share the same p-type substrate as the NMOS

Building CMOS means putting NMOS and PMOS transistors on the very same piece of silicon, even though an NMOS wants a p-type substrate and a PMOS wants an n-type one. The usual solution is an n-well: a pocket of n-type silicon diffused into the p-type substrate, deep enough to host a complete PMOS transistor, source, drain, and body, while the NMOS transistors around it sit directly in the surrounding p-type substrate as before. Everything above that, the oxide layers and the metal wiring that connects the transistors into actual logic gates, is built in the same set of process steps for both transistor types.

1.5.1 Photolithography process

Turning a bare silicon wafer into a working chip is a repeated pattern of deposit, pattern, and etch, applied layer by layer. It starts with the wafer itself, onto which we grow a thin layer of silicon dioxide (SiO2) through oxidation. On top of that, we spin on a photoresist, a light-sensitive polymer. By default we use a negative resist, which becomes soluble wherever light hits it, though a positive resist, which behaves the opposite way, is also an option depending on the process.

Next comes the stepper exposure. A glass mask carrying the pattern for this layer is brought close to the wafer. Its opaque regions block UV light, and its transparent regions let it through, exposing the resist beneath in exactly the shape we want to process. After exposure, we develop and bake the wafer: an acid or base solution washes away the soluble part of the resist, and baking hardens what is left into a solid protective mask.

With that mask in place, acid etching removes material, typically the oxide layer, wherever the resist is not protecting it. A spin, rinse, and dry step cleans the wafer before whatever comes next, whether that is another masked layer or a different process step entirely. Once a layer is finished, we strip away the remaining photoresist, and the wafer is ready for the next pass through this same cycle, one layer at a time, until the full set of transistors and interconnect is complete.

2 Basics of Digital Integrated Circuits

Basics of Digital Integrated Circuits

2.1 Goal for this week

- Classify integrated circuits by fabrication technology (bipolar, MOS/CMOS, BiCMOS) and by scale of integration, and see why Moore's law is the reason those scales keep climbing
- Separate the IC business models: standard parts built for nobody in particular, full-custom ASICs built for one customer, and the semi-custom middle ground
- See how standard cells, macrocells, and gate arrays let an ASIC designer reuse pre-built blocks instead of hand-laying every transistor
- Meet the predecessors to the FPGA: SPLDs (PAL, PLA, PLE) and CPLDs, and the handful of underlying technologies a chip can use to store its configuration

2.2 Digital integrated circuits

A digital integrated circuit (IC) is, in simple terms, a miniature, low-cost circuit made of active and passive elements, all fabricated together on a single silicon die. We can classify ICs by the transistor technology they are built from: bipolar junction transistors (BJT), unipolar MOS or CMOS transistors like the ones we met in the previous chapter, or a mix of both, called BiCMOS.

Bipolar junction transistors have real advantages over CMOS. They switch at higher frequencies, offer better gain, and can drive higher output currents. CMOS wins anyway for almost all digital design, because a BJT's static power consumption is never quite zero, and BJTs are larger and slower to fabricate at a given density. That is exactly the low-static-power, high-density tradeoff that made CMOS the default technology in the previous chapter.

2.2.1 Scale of integration

We also classify ICs by their scale of integration, a rough measure of how many basic elements, transistors or gates, fit on one chip. Small scale integration (SSI) covers up to about 100 elements, medium scale (MSI) 100 to 1,000, large scale (LSI) 1,000 to 10,000, very

large scale (VLSI) 10,000 to 100,000, and ultra large scale (ULSI) 100,000 to 1,000,000. Beyond that, wafer-scale integration treats an entire silicon wafer as a single circuit rather than cutting it into individual dies.

This progression is not just a naming exercise. Gordon Moore observed in 1965 that the number of transistors we can fit on a chip roughly doubles every two years, a trend we now call Moore's law. It is the reason the scale of integration climbed from a handful of gates per chip in the 1960s to transistor counts in the billions today, and it is the main force behind nearly every capability we take for granted in modern digital design.

2.3 Design methodologies

We can also classify digital ICs by who they are designed for, which we call their design methodology: standard ICs (SIC), application-specific ICs (ASIC), and standard ASICs (SASIC).

2.3.1 Standard integrated circuits

A standard IC is designed for an unknown customer. Whoever eventually buys the chip has no influence on how it was designed, and that is exactly what lets it be mass-produced and sold cheaply.

The earliest standard ICs were simple, fixed-function parts, and building any larger system out of them meant wiring together many of them on a board to form whatever circuit was needed. Software-programmable standard parts changed that, above all the CPU: the same piece of hardware adapts to a new task just by changing the program it runs, instead of changing its wiring. That flexibility is not free. A general-purpose CPU is rarely the most efficient way to run any one specific task, a tradeoff we look at more closely when we get to FPGAs later in this book.

2.3.2 Application-specific integrated circuits

An application-specific integrated circuit (ASIC) is designed for one specific application or field, rather than for an unknown general-purpose market. We build ASICs in two ways: full-custom and semi-custom.

2.3.2.1 Full-custom design

A full-custom design is built specifically for one customer, with every mask layer needed to fabricate the chip designed to order for that customer's exact need, whether that is a systolic array, genome analysis hardware, or a dedicated accelerator. The design itself is typically strongly confidential. Since the design cost is comparable to a standard IC's but spread over a far smaller production run, full-custom chips carry a high price per unit.

2.3.2.2 Semi-custom design

Semi-custom design starts from a predesigned or preprocessed chip. Only certain mask layers get adapted to the customer's needs, while the rest stay identical across every customer who buys that base design. The manufacturer keeps a library of pre-tested modules on hand and connects them to fit each customer's requirements, rather than designing from scratch every time. Because the same base design is reused across many customers, semi-custom chips are much cheaper per unit than full-custom ones.

2.3.3 Building blocks for ASIC design

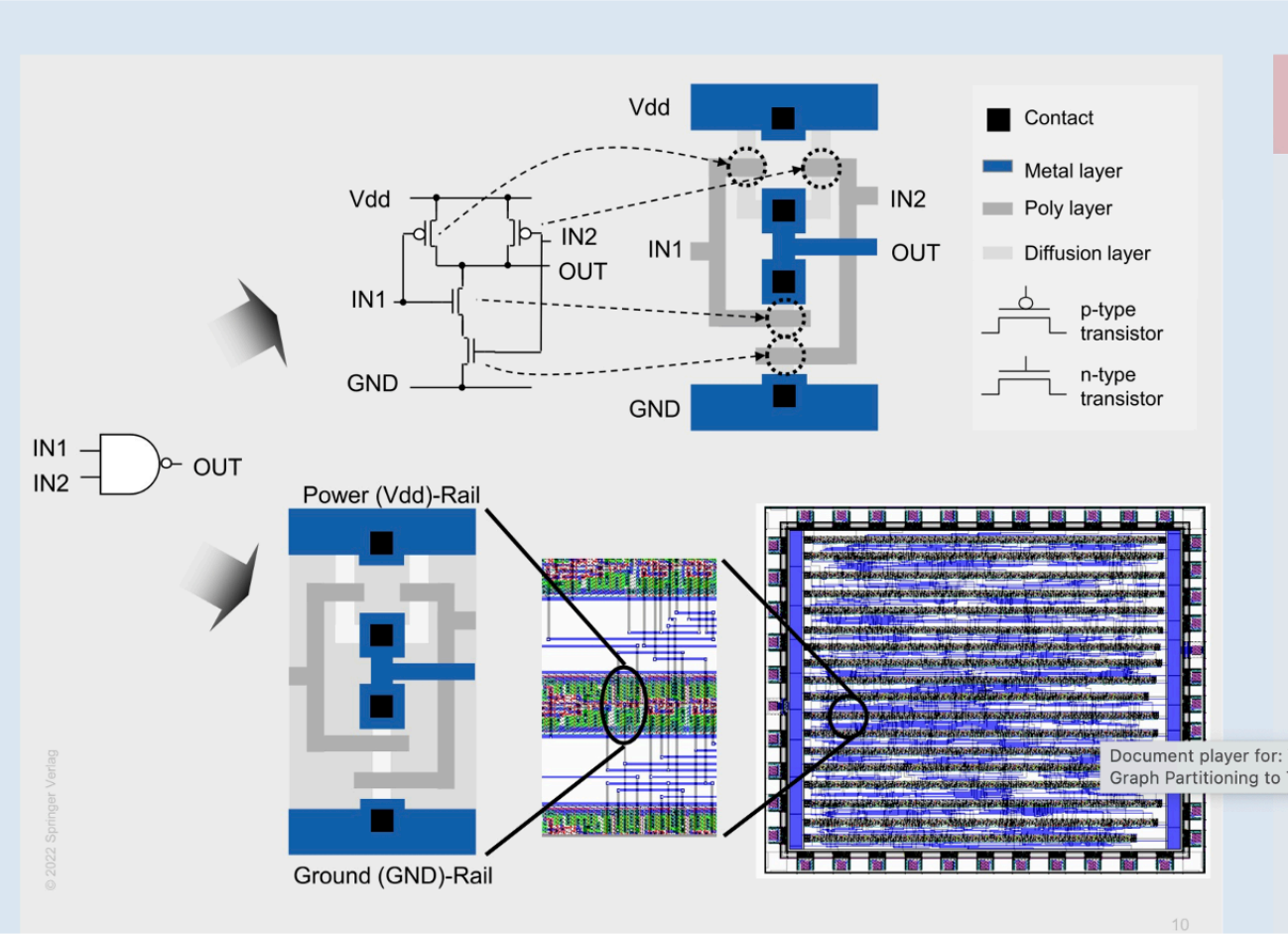


Figure 2.1: Standard-cell design flow: a NAND2 gate as a transistor schematic, its layout in metal, poly, and diffusion layers, and the resulting row-based placement on a die (source: Kahng et al., *VLSI Physical Design: From Graph Partitioning to Timing Closure*, © 2022 Springer Verlag)

Semi-custom and standard-cell ASIC design rests on a few kinds of reusable building block.

Standard cells are small, universal blocks, logic gates, flip-flops, multiplexers, and the like, each with a predefined layout and electrical behavior. The figure above shows a two-input NAND gate as a standard cell: its transistor schematic, its layout in metal, polysilicon, and

diffusion layers, and the fixed-height block that results, ready to sit in a row next to other cells. Every standard cell in a library shares that same row height, which is what lets thousands of them be tiled into rows and wired together with routing that runs above the cells, so-called over-the-cell routing. A chip's entire library of standard cells, together with the data needed to use them, is called a process design kit (PDK).

Macrocells and megacells are a step up from standard cells: larger, pre-laid-out blocks such as adders, multipliers, registers, or even whole microprocessors. The difference between the two is who they were designed for. A megacell is designed without knowledge of the specific application that will use it, which makes it broadly reusable, while a macrocell is designed to order for one customer's needs.

Gate arrays take a different approach. Their basic elements, individual gates or groups of unconnected transistors, are pre-fabricated across the whole die before any customer design is known. Implementing a specific design on a gate array is then purely a metal-wiring problem: connect the gates that are already there in whatever pattern the design needs. Early gate arrays separated these blocks with dedicated routing channels. Later "sea of gates" designs packed transistors edge to edge instead and used higher-level metal layers for routing, improving density, with connections made by shorting or bridging transistors that would otherwise stay unconnected.

2.4 Standard ASICs (SASIC)

A standard ASIC (SASIC), also called a configurable ASIC, is manufactured generically, the same way a standard IC is, and only afterward configured to behave like a custom one. What changes from one application to the next is not the silicon itself. It is a set of configuration bits, held in one of a handful of underlying technologies.

2.4.1 Configuration technologies

Static memory is the technology we will see the most of later in this book. Configuration bits live in SRAM cells and drive electronic switches (transmission gates, tri-state buffers, pass transistors) on or off. It is fast and fully reconfigurable, at the cost of being volatile. This is the same idea behind the configuration SRAM inside every FPGA we meet later in this book.

UV-erasable memory stores its configuration differently. Its key element is a MOSFET with a second gate, electrically isolated and sandwiched between the control gate and the substrate, called a floating gate. A strong lateral field between drain and source injects electrons onto that floating gate, and because it has no conductive path to leak through,

those electrons stay trapped once the field is removed, even with power off. Erasing the device means shining UV light on it, energetic enough to let the trapped electrons escape.

Electrically erasable memory keeps the same floating-gate mechanism, but the device is built so the trapped charge can be removed electrically instead, without needing a UV lamp.

Fuse and antifuse technologies work differently again, and only in one direction. A fuse is a narrow bridge of conducting material that blows open when a programming current is forced through it. An antifuse is the opposite: a thin dielectric separating two conductive layers, which ruptures under a programming voltage and creates a permanent short between them. Either way, programming a fuse-based device is a one-time, permanent decision, unlike the other three technologies, which can all be reconfigured.

2.4.2 Simple programmable logic devices

X	Y	Z	W
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

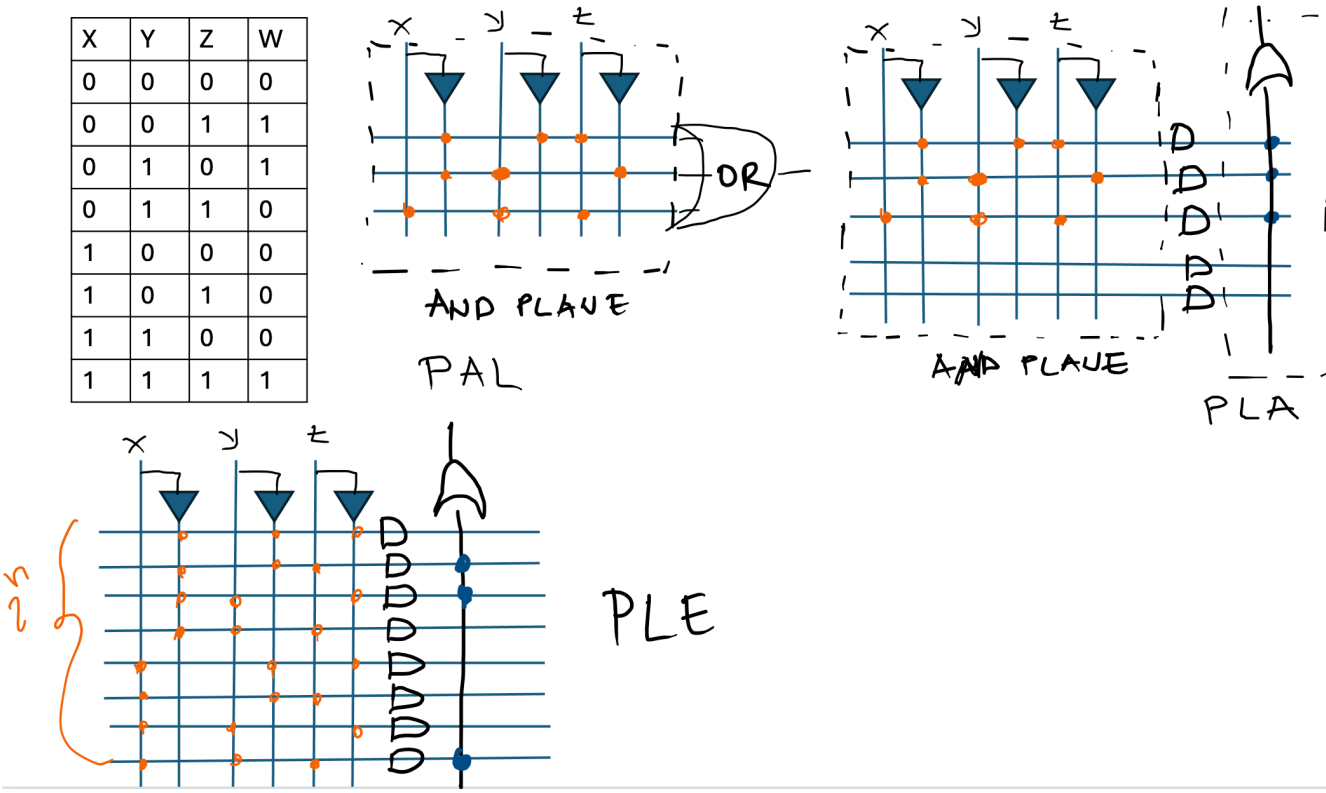


Figure 2.2: SPLD architectures: PAL (configurable AND plane, fixed OR plane), PLA (both planes configurable), and PLE (fixed AND plane covering every product term, configurable OR plane)

Simple programmable logic devices (SPLDs) are the direct ancestors of the FPGA, and the simplest configurable logic devices we will meet. Internally, an SPLD is built from one or two programmable logic planes, an AND plane and an OR plane, that implement a digital function exactly the way we would write it out by hand: as a sum of products. Every input, and its inverted value, runs as a vertical line through the AND plane, and every horizontal line represents one product term. Which vertical lines feed which horizontal line, and which product terms feed which output, is exactly what gets configured.

Which of the two planes is configurable, and which is fixed, gives us three SPLD variants. In a PAL (Programmable Array Logic) device, the AND plane is configurable but the OR plane is fixed. In a PLA (Programmable Logic Array), both planes are configurable, at the cost of more configuration bits and a larger device. In a PLE (Programmable Logic Element), the AND plane is fixed and already generates every possible product term, all 2^n of them for n inputs, and only the OR plane, choosing which of those terms feed each output, is configurable.

2.4.3 Complex programmable logic devices

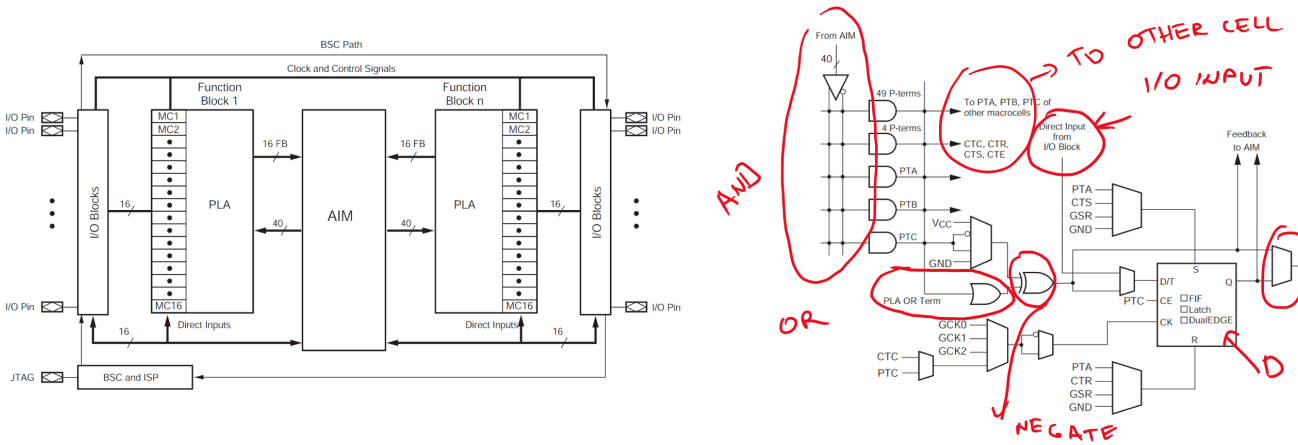


Figure 2.3: CoolRunner-II CPLD architecture: I/O blocks and function blocks tied together by the Advanced Interconnect Matrix (AIM), with one function block's macrocell detail (AND/OR terms, an XOR stage, and a configurable flip-flop) on the right (source: adapted from Xilinx CoolRunner-II documentation, DS090)

A complex programmable logic device (CPLD) is, at its core, many SPLDs packaged on one chip and tied together with a programmable interconnect, trading a single SPLD's limited

capacity for a much larger, still fairly predictable, device.

Take the Xilinx CoolRunner-II as a concrete example. Its I/O blocks connect the chip to the outside world, the same role they play in the FPGAs we meet later in this book. Thirty-two identical function blocks hold the actual logic, and an Advanced Interconnect Matrix (AIM) wires the function blocks to each other and to the I/O blocks, playing the same role the FPGA's routing plane plays later.

Each function block looks like a small PAL: 56 AND product terms feeding 16 OR sum outputs. Beyond that core, each output can pass through an XOR stage, useful for controlled inversion or parity-style logic, and into a flip-flop resource configurable as a D flip-flop, a level-sensitive latch, or a toggle (T) flip-flop. An output mux then picks whether the macrocell's output is the raw combinational result or the registered one, the same combinational-versus-registered choice the FPGA's CLB output mux makes.

Tiling many PAL-like blocks together through the AIM lets a CPLD implement far larger combinational and sequential designs than a single SPLD could, with timing that stays more predictable, and I/O that responds faster, than the larger and more routing-heavy FPGAs we look at later in this book.

3 FPGA Fundamentals

FPGA Fundamentals

3.1 Goal for this week

- Place the FPGA on the spectrum between a general-purpose CPU and a fixed ASIC, and see why that position makes it the right prototyping target for this course
- Break the FPGA fabric down into its three building blocks: CLB, routing plane, I/O blocks
- Open up a CLB far enough to see the LUT + flip-flop pair that will later become “the thing our synthesized SystemVerilog turns into”

3.2 Why reconfigurable hardware?

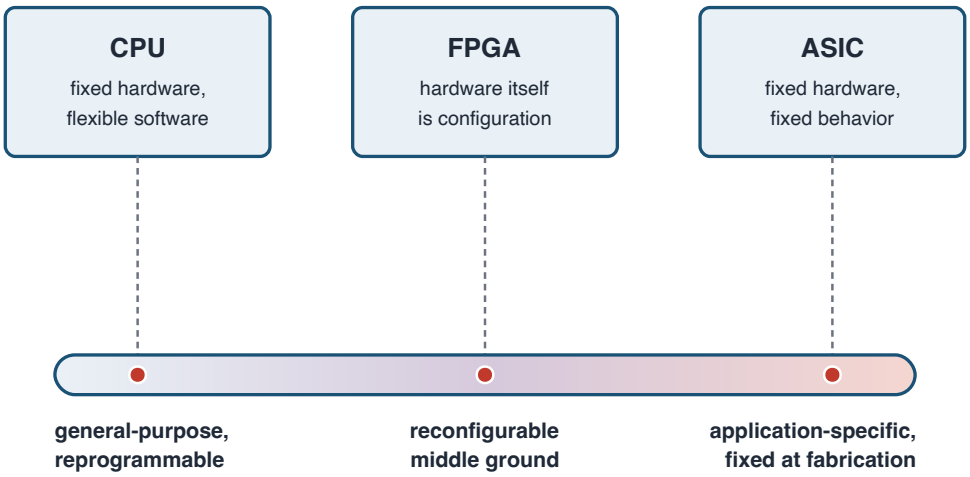


Figure 3.1: The CPU-FPGA-ASIC spectrum: flexibility and reprogrammability trade off against efficiency and fixed-function performance

As we saw in previous chapters, a CPU is a general-purpose integrated circuit: its function is defined not by its hardware but by the software program a developer writes for it. Development cost and time are low, since a single instruction set can be programmed in general-purpose or domain-specific languages to meet almost any demand — from database search to signal processing and beyond. But this same generality is also the CPU's biggest disadvantage. A CPU performs well on workloads like databases, yet it can be slow on compute-intensive problems such as neural network inference.

ASIC designs overcome this general-purpose bottleneck by mapping the application directly into hardware rather than software. A well-known example is Google's Tensor Processing Unit (TPU), built to accelerate the compute-intensive workloads of neural network inference. Its designers tailored the architecture to that one problem, wiring together a matrix of arithmetic logic units (ALUs) instead of following the standard CPU approach of a five-stage pipeline. Comparing a CPU and a TPU on neural network inference, the TPU wins decisively — but hand it a database search instead, and it is useless. This is the core limitation of ASIC accelerators: each one targets a single application, and once fabricated, it cannot be repurposed for anything else.

Comparing the CPU and the TPU, what we would really like is both properties at once: the flexibility to repurpose the device, and hardware tailored to the application at hand to squeeze out its full potential. Reconfigurable hardware, the field-programmable gate array (FPGA), gives us both, and its name spells out how. "Gate array" comes from the internal structure: logic blocks arranged on a matrix-like fabric, echoing the gate-array architecture used in some ASICs. "Programmable" does not mean the FPGA executes a software program in the traditional sense, though it can in a limited way. Instead, it refers to the reconfigurable property itself, achieved by loading a bitstream, a special binary configuration file, onto the device. "Field" refers to the field of application: an FPGA can adapt its internal structure to whatever application is at hand, out in the field, long after manufacturing.

This is exactly the trade-off an FPGA offers. It sits between the CPU and the ASIC by turning the hardware itself into a form of software: the logical function of every building block, and the wiring between them, is set by the bitstream loaded after manufacturing. We get most of an ASIC's advantage of "hardware built for the job" without paying its fixed non-recurring engineering (NRE) cost or its all-or-nothing commitment. That is exactly why this course targets FPGAs. A mistake in a lab assignment costs a re-synthesis and a few minutes, not a re-spun die. FPGAs are also the standard vehicle for rapid prototyping and hardware acceleration in industry for the same reason. Reconfigurability turns hardware design into something you can iterate on.

3.3 The three building blocks

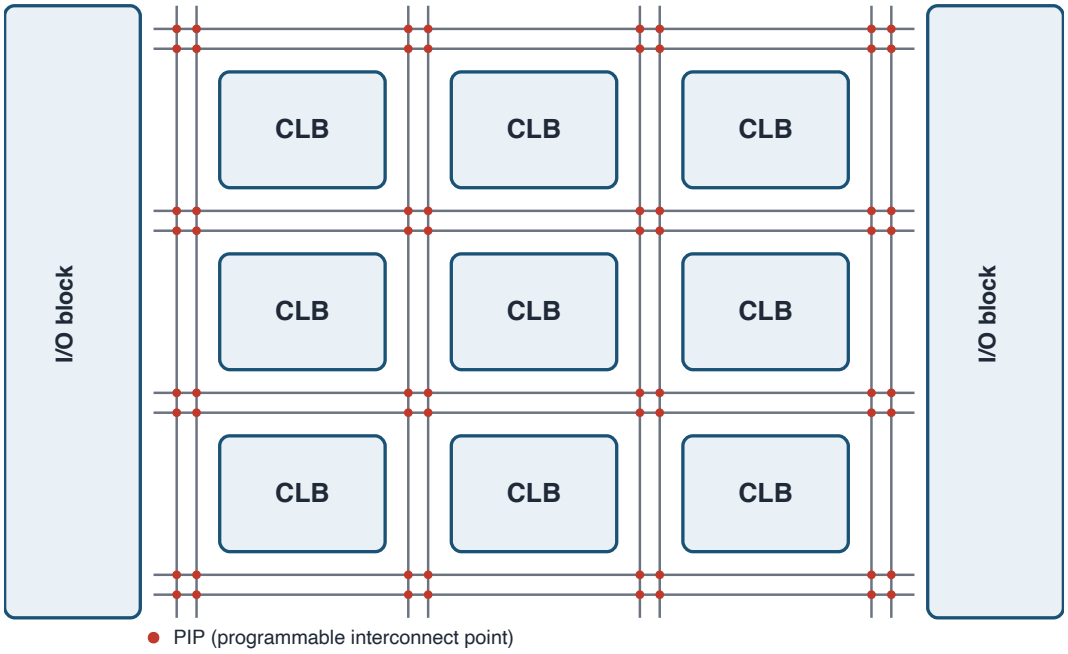


Figure 3.2: FPGA fabric: CLBs embedded in a routing plane, flanked by I/O blocks

Every FPGA, regardless of vendor or generation, is built from the same three kinds of block. CLBs (also called logic elements) sit in a regular grid and implement the actual logic functions and storage of the design. Between and around them runs the routing plane: a fabric of wires and programmable switches whose only job is to connect a CLB's outputs to the inputs of other CLBs, in whatever pattern the design needs. At the edges of the die, I/O blocks connect the fabric to physical pins, translating between the FPGA's internal signal levels and whatever the outside world expects. None of these three blocks is useful without the other two: a CLB with no routing to reach it is dead logic, and routing with nothing to connect is just wire. Placing a design onto an FPGA is precisely the problem of deciding which CLB implements which piece of logic (placement) and which routing resources carry the signals between them (routing). We will meet both steps again when we look at the FPGA design flow.

3.3.1 Configurable Logic Block (CLB)

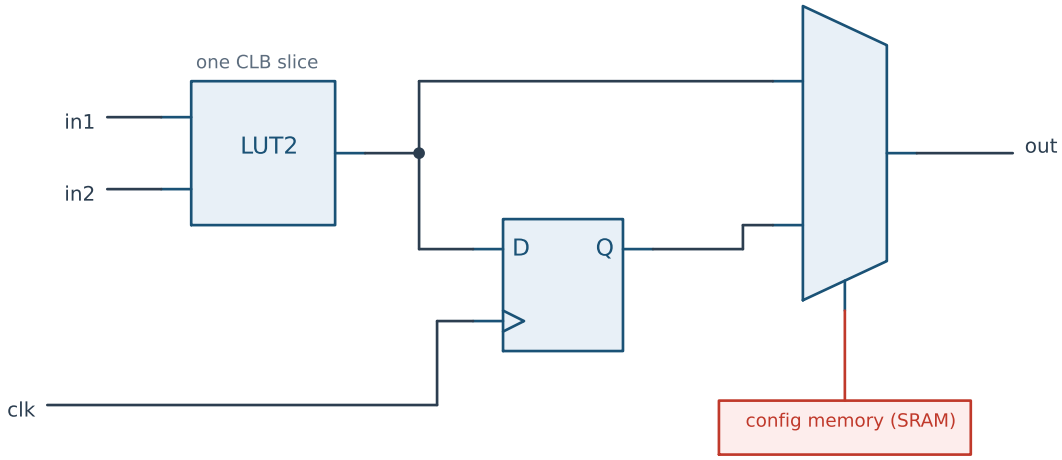


Figure 3.3: One CLB slice: a LUT2 and a flip-flop, with the output mux picking between the combinational and registered result

The CLB is the main carrier of reconfigurability in an FPGA and its elementary building block. You can think of the CLB as the FPGA equivalent of an assembler instruction, the most basic element everything else is built from. The term itself differs across vendors. Xilinx calls it a logic slice, while Altera calls it a logic element. Either way, they refer to the same thing. As the primary building block for digital circuits, a CLB must support combinational logic, sequential logic, and combinations of the two. It consists of a look-up table (LUT), a flip-flop, and an output mux.

A LUT implements combinational logic through memory lookup rather than by wiring gates together. Structurally, it consists of SRAM (static random-access memory) cells paired with a multiplexer that selects which cell's content is forwarded to the output, based on the signal inputs. To see this concretely, consider the smallest useful case, a two-input LUT, or LUT2. With two inputs there are four possible input combinations, so a LUT2 is built from four memory cells and a 4-to-1 multiplexer. The two inputs act as the multiplexer's select lines, in other words as an *address*, and the output is simply whatever bit is stored in the addressed cell.

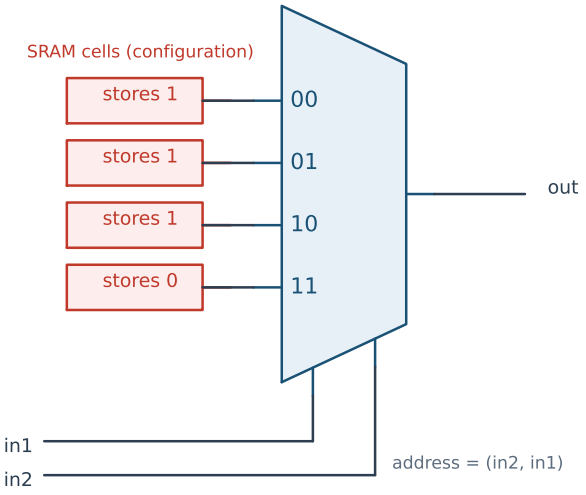


Figure 3.4: LUT2 internals: the two inputs address one of four SRAM cells, and the addressed cell's stored bit is forwarded to the output

To implement logical functions with two input variables, you need to map the truth table into SRAM cells. For example to implement a two-input NAND on a LUT2, we do not build a NAND gate at all: we store the NAND truth table's output column (1, 1, 1, 0) into the four memory cells, in the order addressed by the two inputs, and the LUT now behaves as a NAND for as long as that configuration is loaded. The reconfigurability of LUTs is what makes FPGAs so flexible: the same LUT can just as easily hold an AND, an XOR, or any other two-input function. Reconfiguring the FPGA never touches a wire between gates. It only rewrites these memory cells. Because those cells are ordinary SRAM, they are volatile: power off the board and the "circuit" is gone, which is why every FPGA design needs a configuration step after every power-up.

We do not build larger LUTs as one giant memory array wired to a huge multiplexer. Instead, real FPGA architectures build them recursively from smaller LUTs, adding one extra input and one extra multiplexer stage at a time. A LUT3 is nothing more than two LUT2s, both fed by the same two inputs, with their outputs chosen between by a 2:1 mux controlled by the third input. Extending the same idea, a LUT4 is built from two LUT3s, each in turn built from two LUT2s, combined by one more 2:1 mux controlled by the fourth input. Each additional input doubles the number of underlying LUT2 building blocks and adds one more layer of multiplexing on top. This recursive structure is exactly why vendors describe their larger LUTs as fracturable: the same physical hardware that implements one LUT4 can, with a small configuration change, split back into two independent LUT3s, or four independent LUT2s, letting the synthesis tool pack more, smaller functions into fewer CLBs when a design does not need the full input width.

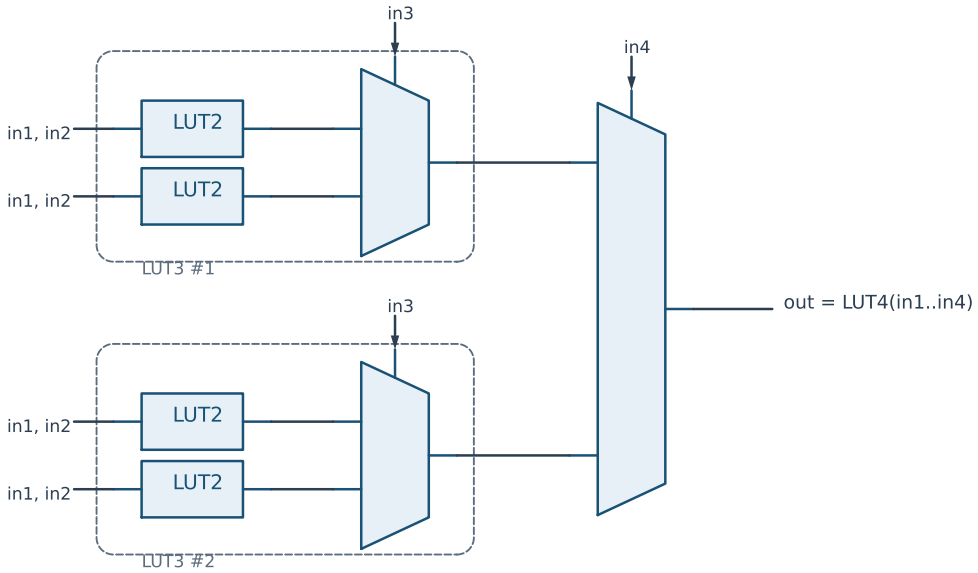


Figure 3.5: Building a LUT4 from four LUT2s: two levels of 2:1 muxing, selected by in3 and in4

Few real digital circuits are purely combinational. Almost every design also needs elements that are synchronous to a clock. To support that, every CLB slice also includes a D flip-flop (DFF) fed from the LUT's output. As we saw earlier, a DFF copies its input (D) to its output (Q) on the active edge of the clock signal, clk. Through configuration bits, the developer can adapt that DFF's behavior, choosing whether it triggers on the rising or falling edge, and setting its initial or reset value. This exposes yet another layer of reconfigurability inside the CLB.

The last element of the CLB is the output mux, which picks between the raw LUT output and the flip-flop's registered output. That mux's select line is, like the LUT's memory cells, just another bit in configuration SRAM. The same physical slice can act as combinational logic or as a single-bit register, purely depending on how it is configured. Real devices bundle several LUT + flip-flop slices, wider LUTs, and extra logic (carry chains, muxes for combining LUTs into wider functions) into one CLB. We look at a concrete example of that when we cover a real device architecture later.

3.3.2 Routing plane: programmable interconnect

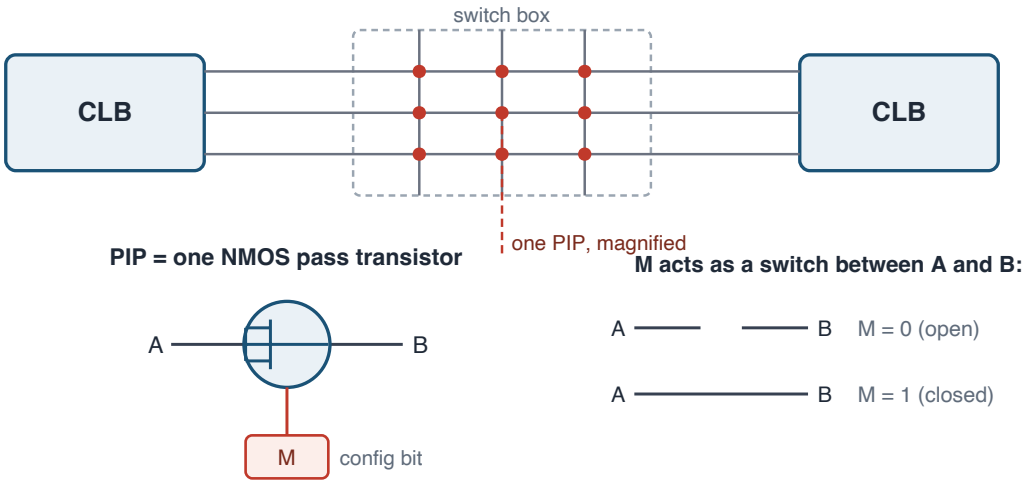


Figure 3.6: Two CLBs connected through a switch box. A PIP is one config-bit-controlled NMOS pass transistor

Routing plays two closely related roles. First, it lets many individual CLBs act as one large digital design: a real circuit needs far more logic and storage than a single CLB slice can hold, so the design is spread across hundreds or thousands of CLBs, and routing is what stitches their individual outputs together into one coherent datapath. Second, routing connects that internal logic to the outside world, joining CLBs to the I/O blocks that carry signals in and out of the fabric. Without it, CLBs and I/O blocks would be correct but disconnected islands of logic.

Between the rows and columns of CLBs run wiring tracks, and at every point where a horizontal and a vertical track cross, a switch box can optionally join them. The atomic component of a switch box is the PIP: a single NMOS transistor used purely as a switch, with its gate driven by one configuration SRAM bit. When that bit is 0, the transistor is off and the two tracks it sits between are electrically separate. When it is 1, the transistor conducts and the tracks become one net. A design's entire wiring, meaning every connection from one CLB's output to another CLB's input, reduces to which of these thousands of PIPs are turned on. This is also why the router is typically the slowest step in the FPGA implementation tools: placement decides which CLB does what, but routing has to find, among a huge and only partially connected mesh of tracks and switch boxes, one specific set of PIPs that realizes every net in the design without any two nets shorting together.

3.3.3 I/O blocks

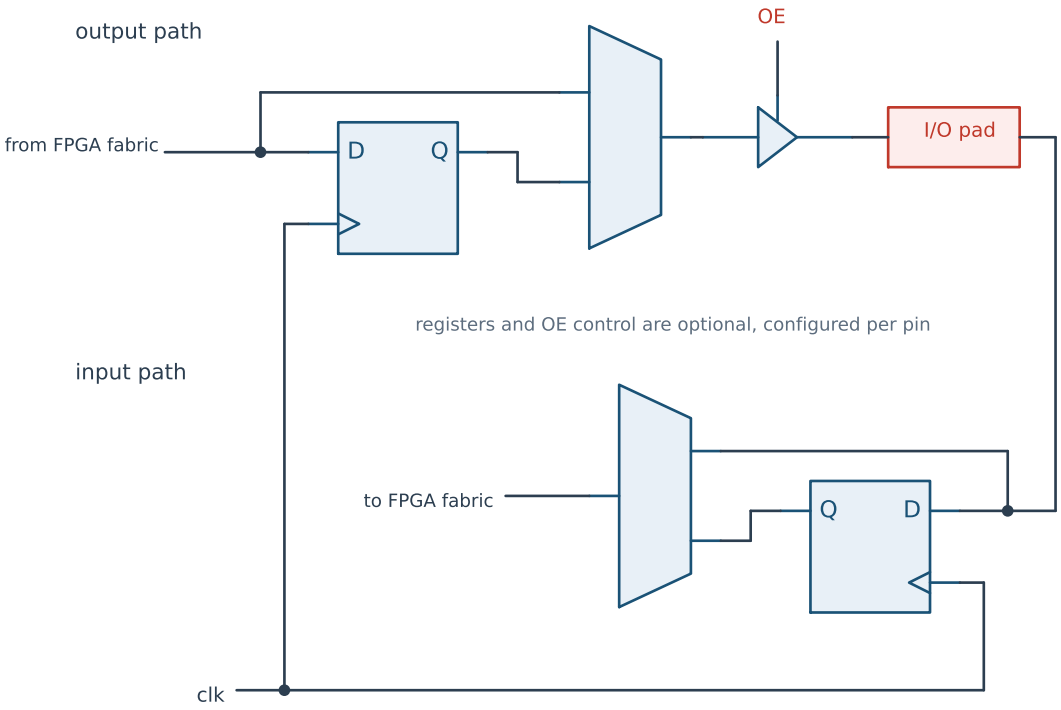


Figure 3.7: I/O block output and input paths: an optional register and a tri-state buffer on output, an optional register on input

An I/O block is the boundary between the reconfigurable fabric and a fixed physical pin, and it has to support more than "just wire a signal through." On the output side, a signal from the fabric can optionally be registered right at the pin before it reaches a tri-state buffer. That buffer's output-enable (OE) input decides whether the pad is actively driven at all, which is what makes a genuinely bidirectional pin possible. The same pad can be released to high-impedance and read as an input whenever the design is not driving it. The input side mirrors this with its own optional register. Registering directly in the I/O block, rather than in the nearest CLB, shortens the path the signal has to travel before it is captured, which matters because I/O timing margins are usually tighter than internal fabric timing. As with everything else on the chip, whether each of these registers is used, and whether a pin behaves as input, output, or bidirectional, is a configuration decision, not a fact fixed at fabrication.

3.3.4 Reconfigurability, level by level

We have now seen five separate places where a single configuration SRAM bit changes what the FPGA does. Inside the CLB, one set of bits picks the function stored in each LUT, another set picks whether the D flip-flop triggers on the rising or falling edge and what value it resets to, and one more bit decides whether the slice's output is the LUT's raw combinational result or the flip-flop's registered one. Out in the routing plane, a bit per PIP decides which pairs of tracks are joined into the same net. At the I/O block, bits decide whether a register is used on the input or output path, and whether a pin drives, receives, or does both. None of these is a special case. Every one of them is the same primitive: an SRAM cell whose value the configuration step writes before the device does anything useful. What looks like five different kinds of reconfigurability is really one mechanism, repeated at every point in the fabric where a design decision has to be made. This is also, in one sentence, what a bitstream is: the full list of values for every one of these bits, for every CLB, PIP, and I/O block on the device.

3.4 Case study: Xilinx Series-7

We have described the CLB, the routing plane, and the I/O block as three abstract building blocks, present in every FPGA regardless of vendor or generation. To see how a real device turns those abstractions into a concrete architecture, we now look at a specific family: Xilinx Series-7. Series-7 comprises four sub-families, Spartan-7, Artix-7, Kintex-7, and Virtex-7, all built from the same underlying fabric while trading off cost, logic capacity, and signal-processing performance. Spartan-7 targets the low end of that range: small, cost-sensitive, high-volume applications. Virtex-7 sits at the opposite end, aimed at the highest connectivity bandwidth, logic capacity, and DSP throughput. Artix-7 and Kintex-7 fill the middle ground. Every member of the family, from the smallest Spartan-7 to the largest Virtex-7, is assembled from the same handful of building blocks described below; only their quantity and mix change from device to device.

3.4.1 CLB: SLICEL and SLICEM

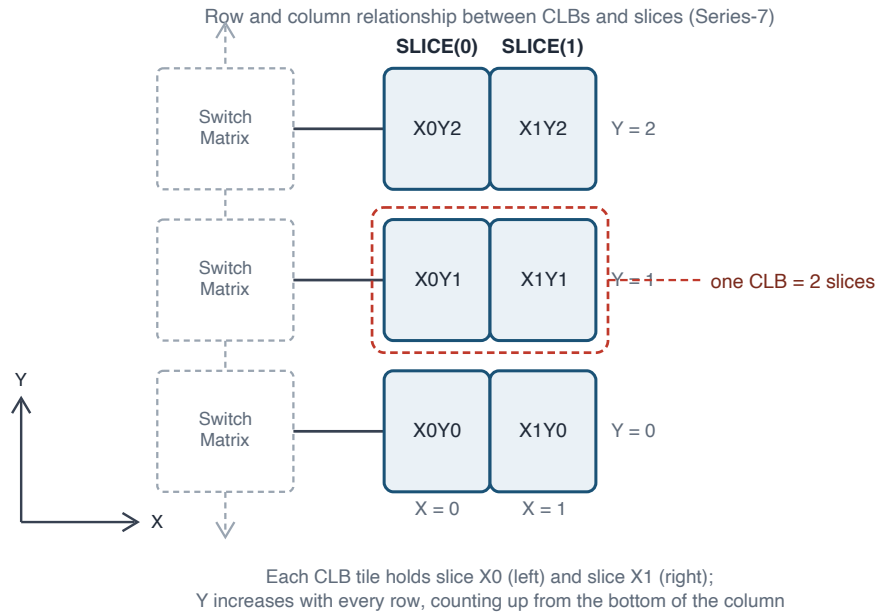


Figure 3.8: Row and column relationship between CLBs and slices: each CLB tile holds slice X0 and slice X1, wired to its switch matrix, with Y counting up from the bottom of the column (redrawn from AMD/Xilinx UG474, Fig. 1)

Where the generic CLB we described earlier held one LUT and one flip-flop, a Series-7 CLB is considerably richer, and Xilinx builds it from two smaller units called slices. Two slice types exist. Every CLB is either a CLB0, containing two SLICEL (“slice logical”) slices, or a CLB1, containing one SLICEM (“slice memory”) and one SLICEL slice. Each slice, of either type, contains four six-input LUTs (LUT6) rather than the single LUT2 we used to introduce the idea — the same recursive fracturable-LUT construction we saw earlier, just taken further, and each LUT6 can equally well operate as two independent LUT5s sharing the same five inputs. To build functions wider than six inputs, the slice adds dedicated combining multiplexers on top of the LUTs themselves: F7AMUX and F7BMUX each merge a pair of LUT6 outputs into one seven-input function, and a further F8MUX merges the two resulting LUT7 outputs into a single eight-input function. This is the same output-mux idea from the generic CLB, just repurposed to widen LUTs instead of choosing between a combinational and a registered output.

Because addition is one of the most common operations in digital design, each slice also includes a dedicated carry chain: fast, purpose-built hardware that propagates carry signals

between adjacent slices without routing them back through the general interconnect, considerably speeding up adders and comparators compared to building the same carry logic out of ordinary LUTs. On the sequential side, each slice packs eight flip-flops rather than the one flip-flop per LUT we introduced earlier. Four of these are configurable to behave as either an edge-triggered flip-flop or a level-sensitive latch, while the remaining four are flip-flop only; all eight share one clock and one clock-enable signal, though each still has its own configuration bits for init value and set/reset behavior, echoing the per-flip-flop reconfigurability we discussed for the generic CLB.

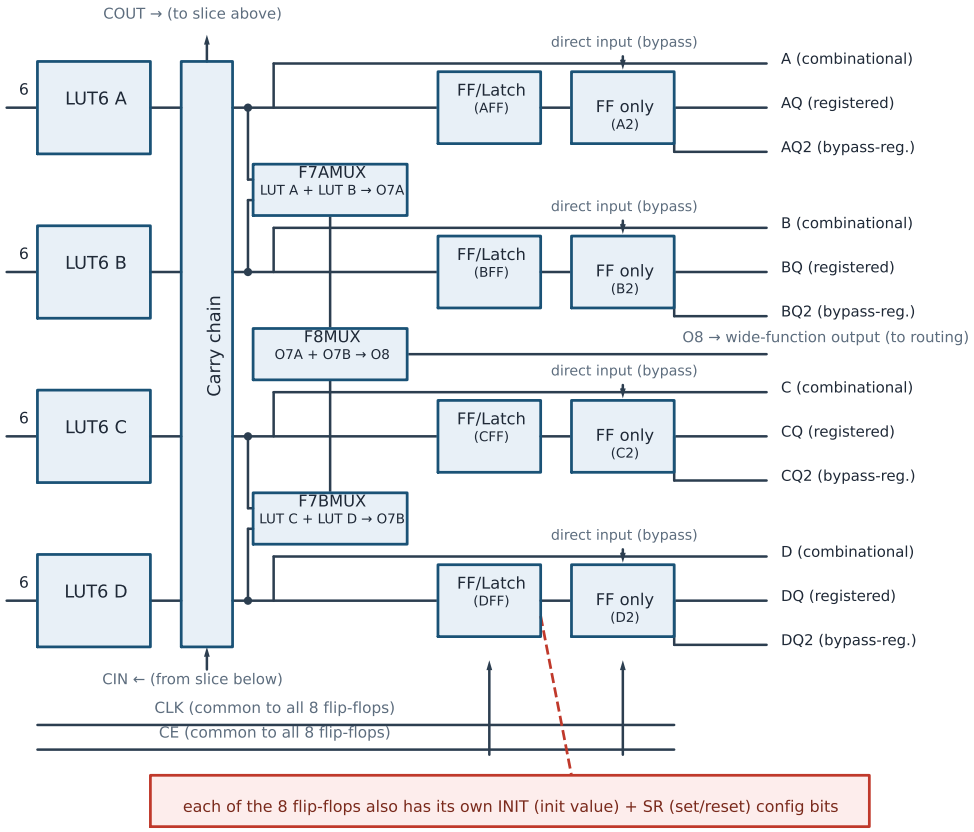


Figure 3.9: SLICEL, simplified block diagram: four LUT6s feed the F7A/F7B/F8 combining muxes and the carry chain, each lane ending in an FF/Latch and an FF-only register sharing one clock and clock-enable (redrawn, not to scale, after AMD/Xilinx UG474, Fig. 2)

SLICEM slices add one more capability that SLICEL slices lack. A LUT's whole purpose, as we saw earlier, is to hold a fixed truth table in SRAM cells that the design's inputs address; that content does not change while the design runs. SLICEM's LUTs can instead be

repurposed as small, fast, genuinely writable memory: either distributed RAM, where the LUT's SRAM cells are written and read directly by the design as if they were a tiny local memory, or a 32-bit shift register, where the same cells behave as a chain of delay elements. Both uses trade away the LUT's normal role as a combinational lookup table in exchange for a small amount of local storage exactly where it is needed, without consuming a dedicated block RAM.

3.4.2 Block RAM and DSP slices

SLICEM's distributed RAM is useful for small amounts of storage, but real designs routinely need much larger memories: register files, FIFOs, line buffers, caches. Building those out of LUTs alone would be both wasteful of logic resources and slow, so Series-7 devices additionally provide dedicated block RAM (BRAM): hard-wired memory macros scattered through the fabric, entirely separate from the CLBs. Each BRAM comes in one of two hard-wired sizes, 36 Kb or 18 Kb, and each size can be configured, at implementation time, into a range of width and depth combinations — a 36 Kb block, for instance, can be organized as 32K words of 1 bit, 16K words of 2 bits, 8K words of 4 bits, and so on, trading width for depth as the design requires. Configuration also selects the memory's port structure: single-port, true dual-port (two fully independent address, read, and write ports), or simple dual-port (one port dedicated to writing, the other to reading). A FIFO is not a separate hardware primitive at all; it is simply a block RAM combined with dedicated pointer and flag logic that turns its linear address space into a first-in-first-out queue.

Arithmetic is common enough, and multiplication expensive enough to build efficiently out of general LUT fabric, that Series-7 devices dedicate a third kind of hard-wired block to it: the DSP slice. A DSP slice implements a fixed pipeline of multiply, add, and accumulate hardware directly in silicon, entirely bypassing the LUT and carry-chain logic used for general-purpose addition. Just as with the CLB and BRAM, a DSP slice is configured rather than wired by hand — configuration bits select which of its internal operations are active and how they are chained together — but the arithmetic itself runs on dedicated hardware rather than being synthesized from LUTs, which is what makes DSP slices dramatically faster and more area-efficient than an equivalent LUT-based multiplier.

3.4.3 Clock management and interconnect

A clock signal is unlike any other net on the chip: every register that uses it must see the same edge at very nearly the same instant, or the design's timing margins erode. General-purpose routing, built from PIPs and switch boxes with no particular guarantee on delay matching, is not good enough for this job, so Series-7 devices route clocks through a dedicated network instead. Global clock buffers distribute a small number of clock signals across the device

with deliberately low skew, and the device's area is partitioned into clock regions, each spanning 50 CLB rows, with its own local clock distribution so that skew stays low even far from the clock's entry point. Before a clock reaches those buffers, it can optionally pass through a clock management tile (CMT), dedicated hardware that synthesizes new frequencies, shifts phase, or removes accumulated skew — capabilities well beyond anything the general fabric could provide.

Everything that is not a dedicated clock net still relies on the same general-purpose interconnect introduced earlier: horizontal and vertical routing tracks running between rows and columns of CLBs, joined at switch boxes wherever they cross. A switch box is simply a cluster of PIPs, and, exactly as in the generic architecture, each PIP reduces to one NMOS pass transistor gated by a single configuration SRAM bit. Series-7 does not introduce a new mechanism here at all — it is the same one-bit-per-connection primitive met earlier, just deployed at the scale a real device needs: thousands of switch boxes, each with many PIPs, giving the router enough freedom to connect any CLB, BRAM, or DSP slice's output to any other block's input.

3.5 FPGA design flow

Earlier, when introducing routing, we noted that placing a design onto an FPGA comes down to two questions: which CLB implements which piece of logic, and which routing resources carry the signals between them. The FPGA design flow is the sequence of tool steps that answers those questions automatically, turning an RTL description into the bitstream that configures every SRAM bit we have discussed in this chapter. That flow closely resembles the one used for ASICs, since both ultimately map a hardware description down to physical resources on silicon, but it differs in one important respect: the target primitive throughout is a LUT and its associated flip-flop, not a standard cell from an ASIC library, which shapes what several of the intermediate steps actually do.

The flow begins with design entry, typically RTL written in a hardware description language, and proceeds to synthesis. Synthesis interprets that RTL and produces a synthesized circuit netlist, generally stored in the standard EDIF format, along with preliminary performance estimates that give an early, optimistic sense of whether the design will meet its timing goals before the slower steps that follow. A translate step then merges this netlist with any others belonging to the same design (for instance, netlists from separately synthesized IP blocks) into one single netlist: a text description of every component in the design and how its terminals connect to every other component's terminals.

The next two steps are where the LUT-based nature of FPGAs really shows. Mapping takes the logical symbols in that merged netlist, ordinary gates such as ANDs, ORs, and multiplexers, and groups them into the physical components the target device actually provides: LUTs of a specific width, flip-flops, carry-chain segments, and so on. This is conceptually

similar to technology mapping in an ASIC flow, except the “cell library” being mapped to is not a fixed set of gates but the fracturable-LUT building blocks we described earlier in this chapter. Placing then takes every one of those mapped components and assigns it to one specific site on the chip, a particular slice within a particular CLB, deciding, in other words, which CLB implements which piece of logic. Routing follows, connecting every placed component to every other one it needs to reach by selecting a path through the switch boxes and PIPs of the routing plane, deciding which routing resources carry each signal.

Routing also produces the reports that tell us whether the design actually works at the clock frequency we want. A static timing analyzer performs a static analysis of the fully placed and routed circuit, accounting for every source of delay along every path, logic delay through LUTs, routing delay through PIPs, and everything in between, to identify the circuit’s critical path: the longest path from the output of one register to the input of another. Because a synchronous design can run no faster than its slowest register-to-register path allows, the critical path directly determines the design’s maximum clock frequency, which is exactly the number this whole flow exists to maximize.

Once placement and routing succeed and timing closes, the last step is bitstream generation: writing out the configuration file for the FPGA, the complete list of values for every LUT content bit, every flip-flop’s configuration, every PIP, and every I/O block setting that we introduced across this chapter. That bit file can be downloaded directly onto the FPGA, configuring its (volatile) SRAM cells as we described earlier, or converted into a PROM file: a non-volatile store that reprograms the FPGA automatically, from power-up, every time the board is powered on.

4 SystemVerilog Basics

SystemVerilog basics

4.1 Goal for this week

- Explain what a hardware description language is, and why we describe digital systems in text instead of drawing every gate
- Place SystemVerilog in the Verilog family, and know which subset of it this course uses
- Read and write a module: ports, directions, internal signals
- Tell nets from variables, and know the four values a signal can hold
- Write numeric constants with an explicit size and base
- Describe combinational logic with `assign` and the SystemVerilog operators, and avoid the classic mistakes: `&` vs. `&&`, and two drivers on one net

4.2 Why describe hardware in text

The previous chapters described digital circuits bottom-up: transistors, gates, and the standard and programmable devices built from them, up to the FPGA. Nobody designs a processor by placing those pieces one at a time. A modern SoC has millions of gates, and a schematic of that size is neither writable nor reviewable. Instead we describe the system in text, in a **hardware description language** (HDL), and let tools do the placing.

The key idea is *abstraction*: an HDL lets us specify how a system operates without getting consumed prematurely by how it is implemented. The same description can be used in two very different ways. A **simulator** runs it as a model, so we can check behavior long before any chip exists. A **synthesizer** turns it into a netlist of real gates, or into the LUTs and flip-flops of the FPGA from the previous chapter. The rest of this course is about writing descriptions that are good for both.

4.3 What an HDL is used for

A single HDL covers a wide range of jobs. At the top, we can write a high-level behavioral model that says *what* a block computes, with no hardware structure at all. One level down

is **register-transfer level** (RTL): the design is a set of registers, and the combinational logic that moves data between them on each clock edge. This is the level at which this course, and almost all digital design, is done. Below it, the output of synthesis is a gate-level netlist, which can itself be written in the same language, and which can be annotated with timing information for timing simulation. Finally, the same language is used to write testbenches that verify the design.

Not everything an HDL can express can be turned into hardware. A simulator will happily run a testbench that waits 10 ns and prints a message, but a synthesizer has no gate for "wait 10 ns". Only a *subset* of the language is synthesizable, and writing in that subset is the first skill of this course.

4.4 From Verilog to SystemVerilog

Verilog was introduced in 1984 with a dual purpose: to model digital systems, and to describe the testbenches that check those models. As we saw, synthesis supports only a subset of what the language can express. Verilog was standardized several times, and the last Verilog standard is the one that SystemVerilog builds on:

Standard	Year	Nickname
IEEE 1364-1995	1995	Verilog-1995
IEEE 1364-2001	2001	Verilog-2001
IEEE 1364.1-2002	2002	Verilog RTL Synthesis (patch for the synthesizable subset)
IEEE 1364-2005	2005	Verilog-2005, the last Verilog standard
IEEE 1800-2005	2005	SystemVerilog-2005, extensions of Verilog-2005
IEEE 1800-2009, -2012, -2017, -2023		SystemVerilog, each extending the previous one

Too many new features had accumulated to keep calling the result Verilog, so a new language was defined. SystemVerilog extends Verilog in two directions: it improves hardware modeling (for example `logic`, `always_comb`, `always_ff`, enumerations and packed structures, all of which we use), and it adds verification features such as assertions, object-oriented programming, and constrained random stimulus.

Each SystemVerilog standard is an extension of the previous one, so backward compatibility is preserved. A synthesizer for SystemVerilog-2005 understands plain Verilog, but a tool

that only supports Verilog-2001 will report an error on the first SystemVerilog construct it meets.

We will cover only a small, synthesizable part of the language. There are two reasons. The language is far too large for one course. And once you can prototype with the basic constructs, new ones are easy to pick up when you need them.

4.5 Anatomy of a module

Every piece of hardware in SystemVerilog lives inside a **module**. The keywords `module` and `endmodule` mark the beginning and the end of the description, and a module can instantiate other modules inside it, which is how hierarchical designs are built.

```
module and_or (
    input  logic a, b, c,
    output logic y
);
  logic t;                 // internal signal
  assign t = a & b;
  assign y = t | c;
endmodule
```

The module name is followed by a **port list** in parentheses. Each port has a *direction*, a *type*, and a name. Several ports of the same direction and type can share one declaration, as `a, b, c` do above. After the semicolon comes the module body: declarations of internal signals such as `t`, and the statements that describe behavior, here two continuous assignments. Figure 4.1 shows what a synthesis tool builds from this module.

Direction	Meaning
`input`	Data flows into the module through this port
`output`	Data flows out of the module through this port
`inout`	Bidirectional: data can flow both ways (used for tri-state pins)

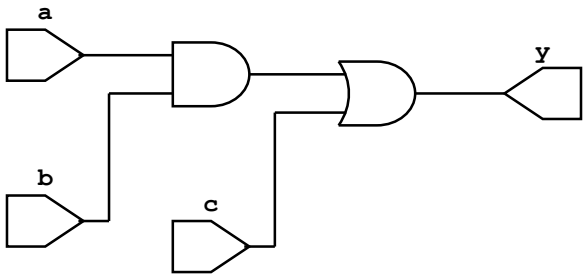


Figure 4.1: The `and_or` module, drawn by Yosys directly from the source above.

4.6 Values: 0, 1, X and Z

A wire in a real circuit is at a high or a low voltage, so you might expect two values, 0 and 1. SystemVerilog uses **four**, because the simulator also has to represent situations that a real circuit can get into:

Value	Meaning
`0`	logic low, false
`1`	logic high, true
`X`	unknown: an uninitialized register, or a net that two sources drive to different values
`Z`	high impedance: nothing is driving the signal (an unconnected input, or a tri-state buffer that is turned off)

You will meet `X` in simulation when something is wrong, and it is one of the most useful debugging signals you have: it spreads through logic that depends on it, so an `X` on an output points back to an uninitialized or double-driven source. `Z` appears with tri-state buffers, like the I/O block of the FPGA chapter. A synthesizer never creates an `X` or a `Z`; they exist only in the simulation model.

4.7 Data types: nets and variables

SystemVerilog has two kinds of signals, and the difference between them is what the whole language is built around.

A **net**, declared with `wire` or `logic`, models a physical connection between things. It stores nothing: its value at any moment is whatever its drivers are producing right now. If several things drive it, the net resolves their values, which is how a tri-state bus works. A **variable** is a signal that *remembers*. It holds the last value assigned to it, and keeps it until the next assignment.

	Net (`wire`,`logic`)	Variable (`logic`, `int`, …)
Models	a connection	a stored value
Holds a value	no, resolved from its drivers	yes, the last one assigned
Driven by	`assign`, module outputs	`assign` (once), or procedural blocks
Multiple drivers	allowed, resolved	an error

In plain Verilog you had to choose between `wire` and `reg` depending on *how you assigned* the signal, and `reg` was widely mistaken for "a register" when it is nothing of the kind. SystemVerilog fixes this with **`logic`**: a four-state data type that can be driven by a continuous assignment, by a procedural block, or by a module output. Its one rule is that it may have only one driver, and the tools check it. For a signal that legitimately has several drivers, such as a tri-state bus, use `wire`.

`bit` is the two-state counterpart of `logic`: it can only hold 0 and 1, so it can hide an unknown value in simulation by silently turning it into a 0. For that reason this course uses `logic` for signals.

4.8 Vectors and integer types

A signal declared with a range in square brackets is a **vector**: `logic [7:0] data` is an 8-bit signal whose most significant bit is `data[7]`. You can select a single bit, `data[3]`, or a contiguous slice, `data[7:4]`. Vectors are unsigned unless declared otherwise.

The types we use most are summarized below. The integer types come from C, and are mainly useful for loop counters and constants. Signals that become wires and registers in hardware should be `logic` vectors, which make the width explicit.

Type	Bits	Signed	States
`logic [n-1:0]`	n	no	4 (0, 1, X, Z)
`bit [n-1:0]`	n	no	2 (0, 1)
`byte`	8	yes	2
`shortint`	16	yes	2

Type	Bits	Signed	States
`int`	32	yes	2
`longint`	64	yes	2
`integer`	32	yes	4 (legacy from Verilog)
`real`	64, floating point	n/a	simulation only: not synthesizable

4.9 Writing constants

A SystemVerilog constant carries its own width, written as `<size>'<base><value>`, where `<size>` is the number of bits and `<base>` is `b`, `o`, `d` or `h` for binary, octal, decimal and hexadecimal. The four constants below all have the value ten, but different widths:

```
logic [3:0] binary_num  = 4'b1010;  // 4-bit binary 1010
logic [5:0] octal_num   = 6'o12;    // 6-bit octal 12
logic [7:0] decimal_num = 8'd10;    // 8-bit decimal 10
logic [7:0] hex_num     = 8'hA;     // 8-bit hexadecimal A
```

Underscores are ignored and are there for readability, so `8'b1010_1010` is easier to read than `8'b10101010`. The characters `x` and `z` are allowed in a constant: `4'b10xz`.

Always write the size. A bare number such as `10` is a **32-bit signed decimal**, and silently mixing it with narrower signals is a common source of width mismatches. If the value is bigger than the stated size, the extra upper bits are discarded: `4'd20` is `4'b0100`. To fill every bit of a signal of any width with the same value, use the unsized forms `'0` and `'1`.

4.10 Continuous assignment

The `assign` keyword connects a value to a signal **continuously**:

```
assign out = a & b;   // out always equals a AND b
```

Whenever `a` or `b` changes, `out` changes with it. A continuous assignment is not a step in a program. It is a gate that exists as long as the circuit is powered, and it models combinational logic.

4.11 Not like assignment in C

A programming language would read the next two lines top to bottom:

```
assign b = a;
assign c = b;
```

First copy `a` into `b`, then copy `b` into `c`. In hardware both lines exist at the same time as two connections, and `b` is just the wire between them. A synthesis tool removes it and infers `assign c = a`. For the same reason, swapping the two lines changes nothing.

4.12 Two drivers on one net

The second example is the one that breaks C intuition completely:

```
wire c;
assign c = a;
assign c = b;
```

In a programming language the second line would overwrite the first, and `c` would end up with the value of `b`. In hardware, both lines are always active, so the net `c` is **driven by two sources at once**. When `a` and `b` agree, nothing is wrong. When they differ, one source pulls the wire high while the other pulls it low, and the simulator reports an unknown value `X` (Figure 4.2). In a real circuit, this is a short circuit between two gate outputs.

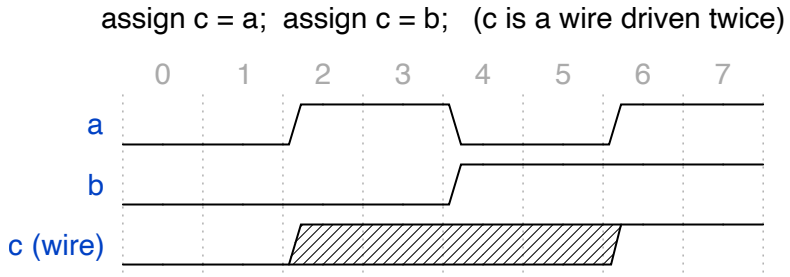


Figure 4.2: A net driven by `a` and by `b`: whenever they disagree, `c` is `X`.

If `c` were declared as `logic`, the tool would reject the second assignment at compile time, because a `logic` signal may have only one driver. That is the reason to prefer `logic`: it turns a silent `X` in simulation into an error you see immediately. This situation must be avoided in every design that is not a deliberate tri-state bus.

4.13 Operators at a glance

SystemVerilog has a rich set of operators, most of them familiar from C. The best way to keep them straight is to group them by *what they return*, because the most common mistakes come from expecting one width and getting another.

Family	Operators	Result
Arithmetic	`+ - * / %`	a number, sized by the surrounding expression
Bitwise	`& | ^ ~`	same width as the operands, computed bit by bit
Reduction	`& | ^ ~& ~| ~^`	1 bit
Logical	`&& || !`	1 bit (true or false)
Relational	`== != < <= > >=`	1 bit
Shift	`<< >> <<< >>>`	width of the left operand
Conditional	`? :`	width of the chosen branch
Concatenation, replication	`{a, b}`, `{n{a}}`	sum of the parts

The sections below take them in turn. Hardware cost matters here: each operator becomes real logic, so `a + b` is an adder and `a * b` is a multiplier, and a division by a non-constant is expensive enough that you should hesitate before writing it.

4.14 Bitwise vs. logical

The bitwise operators `&`, `|`, `^` and `~` work on each bit position separately, and the result is as wide as the operands. The logical operators `&&`, `||` and `!` treat each whole operand as a single truth value (zero is false, anything else is true) and return one bit.

```
logic [3:0] a = 4'b1010;
logic [3:0] b = 4'b0101;

assign x = a & b;     // 4'b0000: no bit position has a 1 in both
assign y = a && b;    // 1'b1: a is non-zero, and so is b
```

Both lines look like “a AND b”, and they give opposite answers. For single-bit signals the two are interchangeable, which is why mixing them up usually goes unnoticed until the signals become vectors. Use `&` and `|` for data, and `&&` and `||` only in conditions.

4.15 Reduction, relational and shift operators

A **reduction** operator is a bitwise operator written with a single operand. It combines all the bits of a vector into one:

```
logic [5:0] a = 6'b101010;
logic c, d, e;

assign c = &a;   // AND of all bits: 0
assign d = |a;   // OR of all bits: 1
assign e = ^a;   // XOR of all bits: 1, the parity of a
```

Reduction operators are a compact way to write a wide gate: `&a` is a six-input AND, and `^a` is a parity tree.

The **relational** operators `== != < <= > >=` compare two values and return one bit. `==` gives `X` if either operand has an `X` or `Z` bit. There is also a case-equality operator, `===`, that compares `X` and `Z` literally, but it exists only for simulation and is not synthesizable. The **shift** operators move the bits of a vector: `<<` and `>>` shift in zeros, and `<<<` and `>>>` are the arithmetic versions, which keep the sign bit of a signed operand when shifting right.

Operator precedence mostly follows C, but you should not rely on remembering it. Add parentheses.

4.16 Conditional operator, concatenation and replication

The **conditional** operator `cond ? x : y` selects between two values. In hardware it is a multiplexer: both `x` and `y` exist, and `cond` chooses which one reaches the output. It can be nested, which is how we build larger multiplexers below.

The **concatenation** operator `{}` joins bits, vectors or expressions into one longer vector. The **replication** operator `{n{...}}` repeats a value `n` times:

```
logic [3:0] a = 4'b1010;
logic [3:0] b = 4'b0101;
logic [7:0] c;
logic [1:0] d = 2'b11;
logic [5:0] e;

assign c = {a, b};     // 8'b10100101
assign e = {3{d}};     // 6'b111111
```

Concatenation is also valid on the left of an assignment, which we use below to capture the carry of an adder.

4.17 Example: 4-bit 4:1 multiplexer

A 4-bit, 4-input multiplexer passes one of four 4-bit inputs to the output, chosen by a 2-bit select signal. Built from nested conditional operators:

```
module mux4to1 (
    input  logic [3:0] in0, in1, in2, in3,   // four 4-bit data inputs
    input  logic [1:0] sel,                  // 2-bit select
    output logic [3:0] out                   // 4-bit output
);
  assign out = (sel == 2'b00) ? in0 :
               (sel == 2'b01) ? in1 :
               (sel == 2'b10) ? in2 :
                                in3;
endmodule
```

Each `?:` tests one value of `sel` and, if it does not match, hands the decision to the next. The last branch, `in3`, is the default that is taken when no earlier condition is true, so there is always a value for `out`. The synthesized result is in Figure 4.3.

4.18 Example: 4-bit adder

The adder adds two 4-bit numbers and produces a 4-bit sum and a carry-out bit. We build it exactly as on paper: a half adder for the least significant bit, then a full adder for each following bit, with an internal carry wire `c1`, `c2`, `c3` passed from one stage to the next.

```
module adder4bit (
    input  logic [3:0] a, b,
    output logic [3:0] sum,
    output logic       carry_out
);
  logic c1, c2, c3;                      // carries between stages

  // Half adder: least significant bit
  assign sum[0] = a[0] ^ b[0];
```

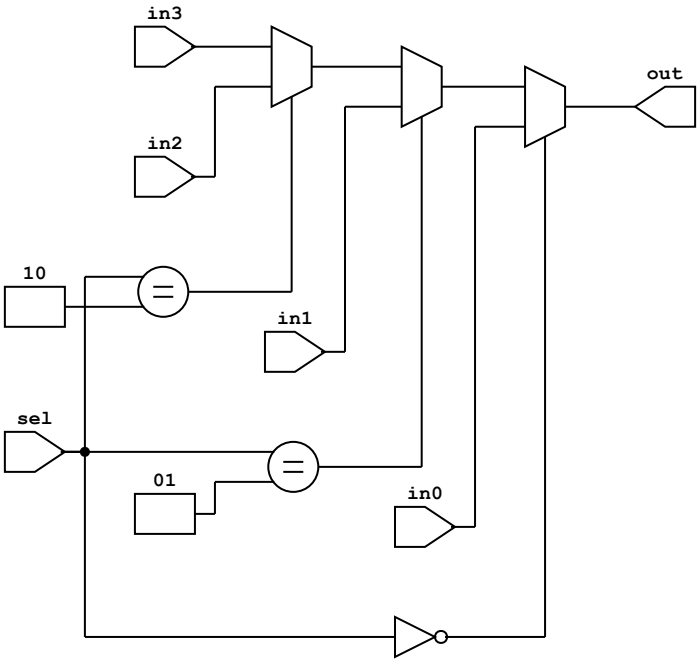


Figure 4.3: The `mux4to1` module as drawn by Yosys.

```
  assign c1     = a[0] & b[0];

  // Full adders
  assign sum[1] = a[1] ^ b[1] ^ c1;
  assign c2     = (a[1] & b[1]) | (c1 & (a[1] ^ b[1]));

  assign sum[2] = a[2] ^ b[2] ^ c2;
  assign c3     = (a[2] & b[2]) | (c2 & (a[2] ^ b[2]));

  assign sum[3]    = a[3] ^ b[3] ^ c3;
  assign carry_out = (a[3] & b[3]) | (c3 & (a[3] ^ b[3]));
endmodule
```

Writing out the gate equations is useful once, to see what an adder is. In practice we let the tool do it. The + operator, with a concatenation on the left to catch the carry, describes the same adder in one line:

```
assign {carry_out, sum} = a + b;
```

The 5-bit result of `a + b` is split across the 1-bit `carry_out` and the 4-bit `sum`. Figure 4.4 shows the hand-built carry chain, and Figure 4.5 the single adder block that a synthesis tool is free to map onto the dedicated carry logic of an FPGA.

4.19 From modules to behavior

Everything we wrote in this chapter is combinational: outputs depend only on the current inputs, and nothing remembers anything. That was enough to describe multiplexers and adders, but a processor also needs state. That calls for procedural blocks, which the next chapter introduces: `always_comb` for larger combinational descriptions, and `always_ff` for registers that update on a clock edge.

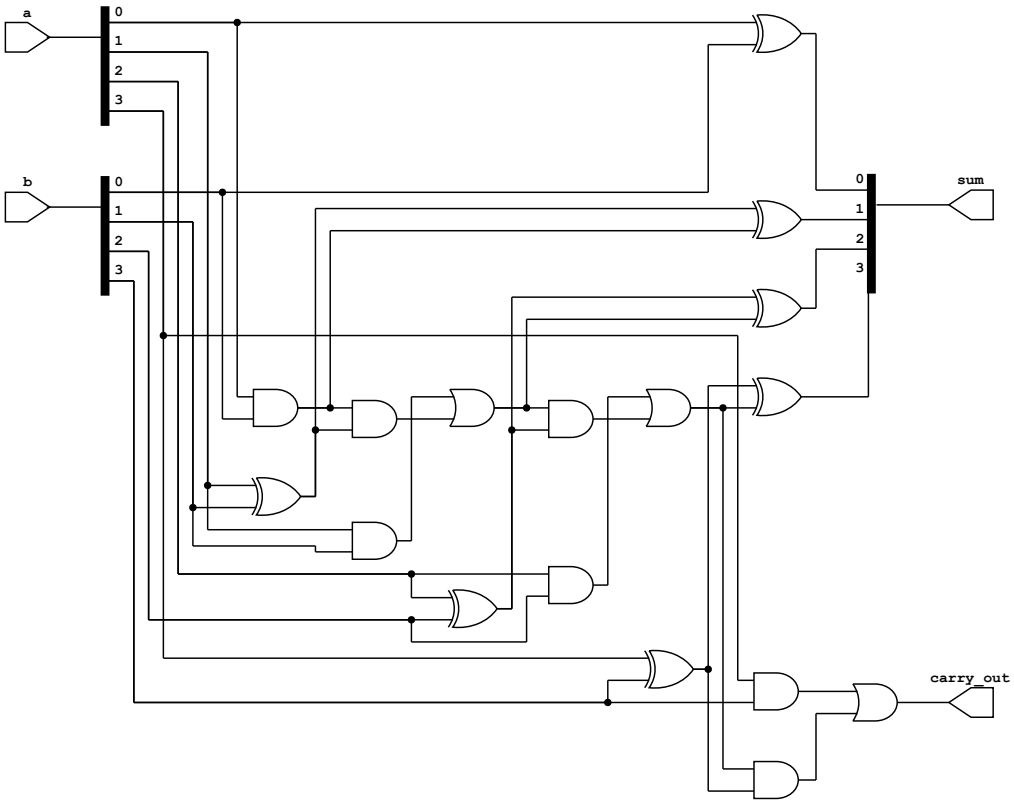


Figure 4.4: The gate-level `adder4bit`: a half adder followed by three full adders, with the carry rippling from bit to bit.

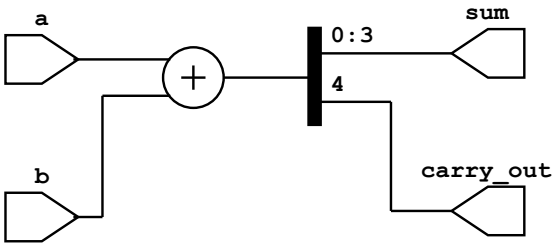


Figure 4.5: The same adder written with the + operator.

5 Designing with SystemVerilog

Designing with SystemVerilog

5.1 Goal for this week

- Tell the two styles of hardware description apart: behavioral and structural
- Describe sequential logic with `always_ff`: flip-flops, resets, registers with an enable, and counters
- Know what a latch is, and write code that cannot produce one by accident
- Describe combinational logic with `always_comb`: multiplexers and decoders
- Explain the difference between `=` and `<=`, and why each block uses one of them
- Assemble a design from smaller modules, and test it with a testbench

5.2 Two styles of design

There are two ways to describe a piece of hardware in SystemVerilog, and a real design uses both.

In **behavioral** design we describe *what* a block does: its outputs as a function of its inputs and, for sequential logic, of its stored state. We write that with `assign` and with `always` blocks, and leave it to the synthesis tool to pick the gates and flip-flops. In **structural** design we describe *how* a block is composed of other blocks: we create instances of smaller modules and connect their ports with wires, much as one would wire chips on a board.

	Behavioral	Structural
Describes	**what** a block does	**how** it is built from other blocks
Written with	`assign`, `always_*`	module **instances**
Example	a register, a multiplexer	a pipeline stage

The two styles meet at the leaf cells. In the HIPV SoC that we use as a running example, the folder `rtl/generic` holds small, reusable cells (`flopenr`, `mux2`, `mux3`, `adder`) written in behavioral style, and the larger modules of the processor instantiate them structurally. This

chapter begins with the behavioral style, because those leaf cells have to be written before they can be assembled, and ends with the structural style.

5.3 The `always` block

The previous chapter described logic one line at a time with `assign`. That is enough for an adder, but awkward the moment a function needs an `if` or a `case`. For those we use an **`always` block**: a block of statements that describes how a set of outputs depends on a set of inputs.

SystemVerilog has three kinds of `always` block for hardware, and which one you use is dictated by what you are building:

Block	Describes	Runs when
`always_comb`	combinational logic	any signal it **reads** changes
`always_ff @(posedge clk)`	flip-flops	the clock edge
`always_latch`	latches	any of its inputs changes

The names are a promise to the tool: if the code inside does not describe the kind of hardware the block name claims, the tool reports it. Older Verilog has a single `always @(...)` for all three purposes, and a whole class of bugs came from getting it wrong without any warning.

Two facts about `always` blocks are worth keeping in mind from the start. First, every `always` block in a module is a separate piece of hardware that exists all the time, like an `assign`, so they all run *concurrently* and the order in which you write them does not matter. Second, the statements *inside* one block run in the order written, like a small program. A block is a description of hardware, not a function that gets called.

5.4 Sequential logic: describe the hardware you want

Hardware description is not programming, and the habit that separates the two is this: **decide on the hardware first, then write the code that describes it.** If you cannot sketch the flip-flops and gates you expect, you are not ready to write the `always` block, and the synthesis tool will build something, but not necessarily what you meant.

So for every circuit in this chapter we follow the same four steps. First, the hardware we want. Second, its **waveform**, which says what the circuit must do over time and is the most precise specification there is. Third, the code that describes it. Fourth, a walk through the

waveform to check that the code matches it. Learning to read waveforms is half of the work: it is also how you will debug in simulation.

A convention for all the diagrams in this chapter: the clock rises at the start of each numbered column, and input signals change in the *middle* of a column, after the clock has fallen. That way an input never changes at the same instant as a rising edge, and it is always clear which value the circuit sampled.

5.5 The D flip-flop

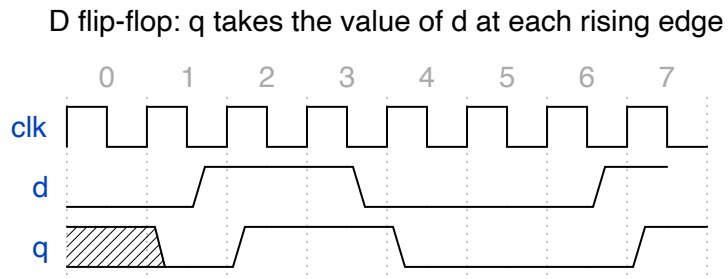


Figure 5.1: A D flip-flop: `q` takes the value of `d` at each rising clock edge.

The flip-flop is the memory element of synchronous design. It has a data input `d`, a clock `clk` and an output `q`. At every rising edge of the clock it copies `d` to `q`, and between edges it holds that value no matter what `d` does (Figure 5.1).

Follow the waveform from left to right. In column 0, `q` is unknown (the hatched area): nothing has been stored yet. At the rising edge that starts column 1, `d` is 0, so `q` becomes 0. In the middle of column 1, `d` rises, and `q` does **not** react: the flip-flop is not looking. At the next rising edge, starting column 2, it samples `d = 1` and `q` rises. The same happens in reverse when `d` falls in the middle of column 3: `q` falls only at the edge that starts column 4. A flip-flop is a circuit that looks at its input at one instant per cycle.

The code is as short as the circuit:

```
module dff (
    input  logic clk, d,
    output logic q
);
  always_ff @(posedge clk) begin
    q <= d;
```

```
  end
endmodule
```

The sensitivity list `@(posedge clk)` names the event that makes the block run, and the block runs *only* then. Between clock edges nothing happens and `q` keeps its value: that holding is the memory. The operator is `<=`, the non-blocking assignment, which we explain at the end of the chapter; for now, remember that it goes with `always_ff`.

5.6 The latch

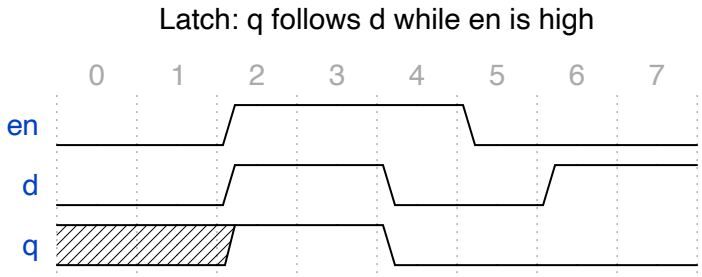


Figure 5.2: A latch: `q` follows `d` while `en` is high, and holds its value while `en` is low.

A **latch** is a memory element too, but it looks at its input for a whole *level*, not at one instant. While the enable `en` is high, the latch is *transparent*: `q` follows `d` continuously. When `en` falls, the latch closes and `q` holds the last value (Figure 5.2).

In the waveform, `en` is high during columns 2 to 4. As soon as it rises, `q` takes the value of `d`, which is 1. When `d` falls at the start of column 4, `q` falls with it, because the latch is still open. `en` falls at the start of column 5, and from then on `q` stays 0: when `d` rises again in column 6, the latch is closed and does not notice. Compare this with the flip-flop of the previous waveform, where `q` could only change at a clock edge.

```
always_latch begin
  if (en) q <= d;
end
```

A latch has no clock edge, so a signal passing through it can ripple along for as long as `en` is high, which makes timing analysis difficult. In a synchronous design we want flip-flops everywhere, and we use a latch only on purpose. The reason to know it exists is that it is easy to build one *by accident*, and the section on combinational logic below shows how.

5.7 Adding a reset

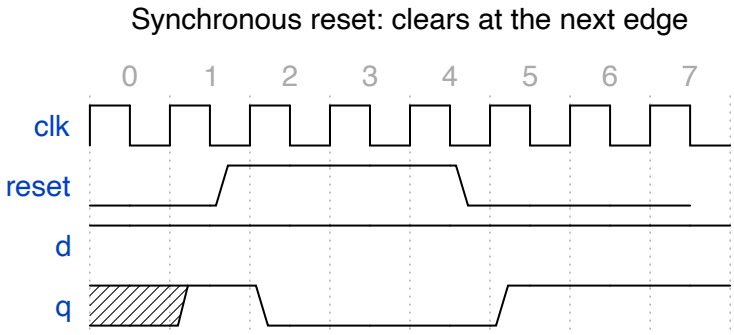


Figure 5.3: Synchronous reset: `q` clears at the next rising edge after `reset` rises.

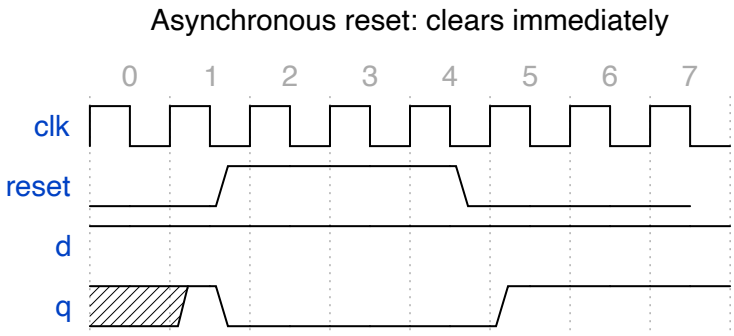


Figure 5.4: Asynchronous reset: `q` clears the moment `reset` rises, without waiting for a clock edge.

A flip-flop that powers up with an unknown value is a bug waiting to happen, so registers get a **reset** that forces a known value. There are two ways to write it, and both clear `q` when `reset` is high. They differ in *when* the clear takes effect.

With a **synchronous** reset (Figure 5.3), `reset` is just another input that the flip-flop looks at on the clock edge. In the waveform `d` stays 1. `q` becomes 1 at the edge that starts column 1. `reset` rises in the middle of column 1 and `q` does not move; only at the edge starting column 2 does the flip-flop see `reset` high and clear. It stays cleared through the edges of columns 3 and 4 because `reset` is still high. `reset` falls in the middle of column 4, and at the edge starting column 5 the flip-flop loads `d` again.

```
always_ff @(posedge clk) begin
  if (reset) q <= 1'b0;
  else       q <= d;
end
```

With an **asynchronous** reset (Figure 5.4), the flip-flop reacts to `reset` itself, whatever the clock is doing. The same stimulus now clears `q` in the *middle* of column 1, the instant `reset` rises, not at the next edge. It stays 0 while `reset` is high, and after `reset` falls it waits for the next edge, at the start of column 5, to load `d`.

```
always_ff @(posedge clk or posedge reset) begin
  if (reset) q <= 1'b0;
  else       q <= d;
end
```

This is why `reset` appears in the sensitivity list next to `posedge clk`: the block must also run when `reset` rises. An asynchronous reset needs a flip-flop with a dedicated reset pin, and its release matters: if `reset` falls too close to a clock edge the flip-flop can misbehave, a problem we return to when we discuss synchronizers. A synchronous reset needs a running clock, but is just logic in front of the D input (or a dedicated set/reset input on an FPGA flip-flop, like the one in the slice of the previous chapter).

The HIPV SoC chooses one style for the whole design: its coding standard requires a **synchronous** reset throughout. That is the style we use from now on. Pick one style, use it everywhere, and know which one you picked.

5.8 Registers and the enable signal

A single flip-flop stores one bit. A **register** is a group of flip-flops that load together, and in a processor most registers also need an **enable**: a signal that says "load a new value at this edge", and otherwise "keep the old one". A pipeline register that must hold its contents while the stage ahead is busy is exactly that. This is the most-instantiated cell in HIPV:

```
module flopenr #(parameter WIDTH = 8) (
  input  logic             clk, reset, en,
  input  logic [WIDTH-1:0] D,
  output logic [WIDTH-1:0] Q);

  always_ff @(posedge clk)
    if (reset) Q <= '0;
```

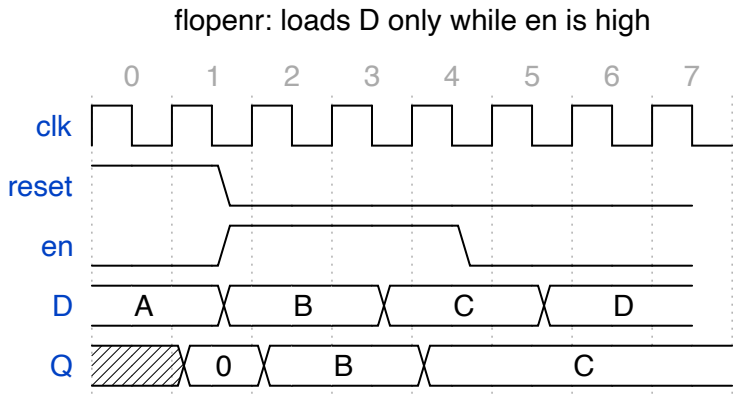


Figure 5.5: `flopenr`: after the reset, `Q` loads `D` only at the edges where `en` is high.

```
    else if (en) Q <= D;

endmodule
```

The module is the D flip-flop with a synchronous reset, plus one more condition. `WIDTH` is a parameter, so the same description serves for a 1-bit flag and for the 32-bit program counter. The priority is spelled out by the `if`/`else if` chain: reset wins, then enable, and if neither is active there is no assignment at all, so the register keeps its value. That "no assignment means hold" is legal inside `always_ff`, since a flip-flop naturally remembers; it is the opposite of the rule for `always_comb`, where it would build a latch.

Figure 5.5 shows the behavior. `reset` is high around the first edge, so at the start of column 1 `Q` loads 0. `D` carries the values A, B, C, D, changing mid-column. `en` is high for the edges starting columns 2, 3 and 4. At column 2 the register loads B. At column 3 `D` is still B, so `Q` stays B. At column 4 `D` is C, so `Q` loads C. Then `en` falls in the middle of column 4, and from the edge at column 5 on the register ignores `D`, even though `D` keeps changing: `Q` holds C. Figure 5.6 is what the synthesis tool builds: one flip-flop cell with clock, data, enable and synchronous reset inputs, which is what an FPGA slice flip-flop provides directly (the clock-enable and set/reset inputs of the previous chapter).

5.9 Counters: two ways to write one

A counter is a register whose next value is its own current value plus one. Its waveform is the simplest of any sequential circuit: after the reset, the output goes up by one at every

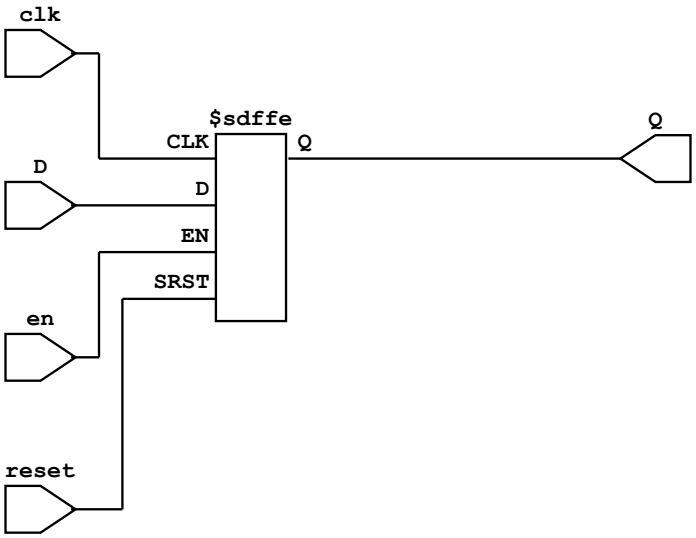


Figure 5.6: The `flopenr` module (8-bit by default), drawn by Yosys from the real HIPV file. `$sdffe` is a flip-flop with **s**ynchronous reset and **e**nable.

clock edge (Figure 5.7). There are two common ways to write it, and the best habit is to know both.

Way 1: everything in one `always_ff`. The straightforward form says, at each edge, "if reset, clear; otherwise `count <= count + 1`":

```
module counter4 (
    input  logic       clk,
    input  logic       reset,
    output logic [3:0] count
);
  always_ff @(posedge clk) begin
    if (reset) count <= 4'b0000;
    else       count <= count + 1;
  end
endmodule
```

At the edge starting column 1, `reset` is still high and `count` loads 0. From then on `reset` is low, and at each edge `count` takes its own value plus one: 1, 2, 3 and so on. When it reaches 15, the next `+ 1` wraps around to 0, because the result is truncated to 4 bits.

Way 2: next-state logic and a register. The second form splits the description in two.

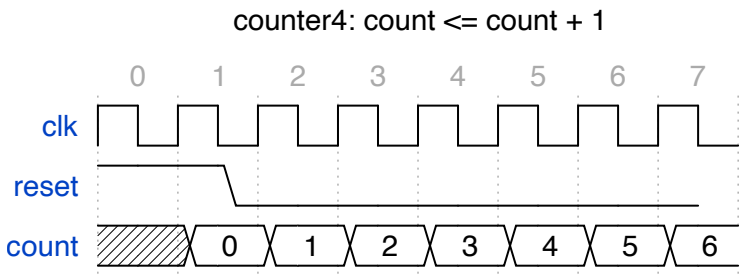


Figure 5.7: Timing of `counter4`: after the reset, `count` increases by one at every rising edge.

A combinational block computes what the next value *would be*, `count_next`, and a plain register stores it:

```
module counter4_next (
    input  logic       clk,
    input  logic       reset,
    output logic [3:0] count
);
  logic [3:0] count_next;

  always_comb begin
    count_next = count + 1;
  end

  always_ff @(posedge clk) begin
    if (reset) count <= 4'b0000;
    else       count <= count_next;
  end
endmodule
```

The waveform now has one more signal (Figure 5.8). `count_next` is combinational, so it changes as soon as `count` does, and is always one higher. In column 1, `count` is 0 and `count_next` is already 1, but the register ignores it at the edge starting column 1 because `reset` is high. At the edge starting column 2 the register copies the 1, and so on.

Both modules synthesize to exactly the same hardware: an adder, a register with a synchronous reset, and the feedback wire (Figure 5.9).

So why bother with the second form? Because it separates two questions: *what is the next value?* (combinational logic) and *when do we store it?* (the register). For a counter the

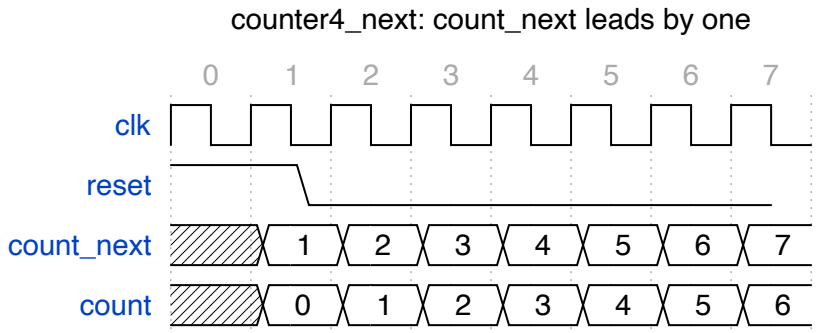


Figure 5.8: Timing of `counter4_next`: `count_next` is always one ahead of `count`, and the register copies it at each edge.

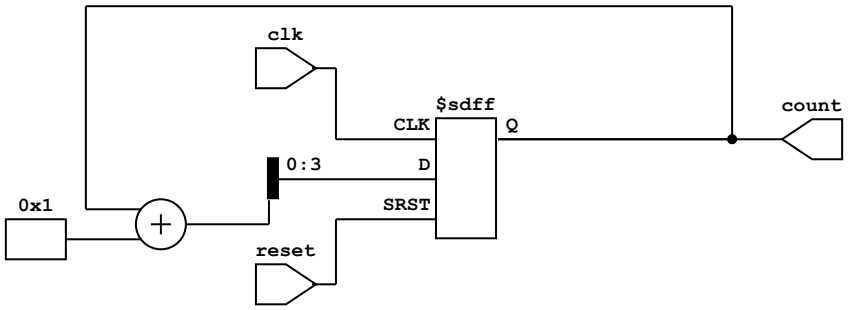


Figure 5.9: Both counter descriptions give this circuit, drawn by Yosys: the register output feeds the output port and the adder, and `reset` drives the synchronous reset input.

answer to the first is trivial, and way 1 is perfectly fine. For anything bigger, such as a state machine or a pipeline stage, the next-state logic can be dozens of lines of `if` and `case`, and keeping it out of the clocked block makes it far easier to read and to check. The HIPV SoC uses this idea in its 64-bit timer counter, `cntr`, which computes `plus_one = count + 1` with an `assign` and feeds it to a register cell with an enable and a clear, instead of writing the increment inside the clocked block.

5.10 Combinational logic

We now do the same for combinational circuits. The method is identical: hardware, waveform, code, walk through. The difference is in the waveform: there is no clock, because a combinational output depends only on the current inputs, so it changes as soon as an input does.

The block for it is `always_comb`. It re-evaluates whenever any signal it reads changes. It is closer to a spreadsheet formula that recalculates on every input edit than to a function you call. You write no sensitivity list, since the tool works it out from the signals the block reads. The assignment inside is the blocking `=`, which reads as “this is what the output equals right now”. There is no state to remember, which gives one rule that matters more than any other: **every output must be assigned on every path through the block**.

5.11 The multiplexer

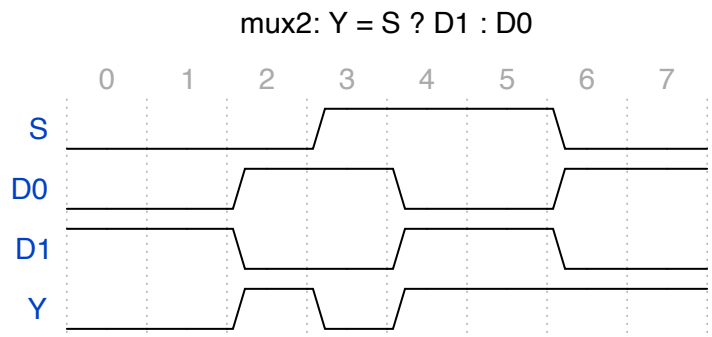


Figure 5.10: A 2-to-1 multiplexer: `Y` follows `D0` while `S` is 0 and `D1` while `S` is 1, with no clock.

A multiplexer selects one of several inputs. The 2-to-1 version is the first cell in `rtl/generic`, and HIPV writes it as a continuous assignment:

```
module mux2 #(parameter WIDTH = 8) (
  input  logic [WIDTH-1:0] D0, D1,
  input  logic             S,
  output logic [WIDTH-1:0] Y);

  assign Y = S ? D1 : D0;

endmodule
```

Figure 5.10 shows what it does. There is no clock: `Y` equals `D0` while `S` is 0 and `D1` while `S` is 1, and it changes the moment the selected input or the select changes. At the start of column 2 `D0` rises, and since `S` is 0, so does `Y`. At the start of column 3 `S` rises, `Y` switches over to `D1`, which is 0, and falls. At column 4 `D1` rises, `Y` follows. At column 6 `S` falls and `Y` returns to `D0`, which is 1.

The same multiplexer in an `always_comb` block reads like this:

```
always_comb begin
  if (S) Y = D1;
  else   Y = D0;
end
```

Both give the same circuit. The `assign` form is shorter for one line of logic; the `always_comb` form is better when there are several outputs or several conditions. For a multiplexer with more than two inputs, the `case` statement is clearer than a chain of `?:` operators. HIPV's 3-to-1 multiplexer, written with a chain of conditional operators in the repository, looks like this as an `always_comb` with `case`:

```
module mux3 #(parameter WIDTH = 8) (
  input  logic [WIDTH-1:0] D0, D1, D2,
  input  logic [1:0]       S,
  output logic [WIDTH-1:0] Y);

  always_comb begin
    case (S)
      2'b01:   Y = D1;
      2'b10:   Y = D2;
      default: Y = D0;    // 2'b00, and 2'b11 falls back to D0
    endcase
  end
endmodule
```

The `case` compares `S` against each listed value and runs the matching branch. The **`default`** branch is not optional: it makes sure `Y` is assigned for *every* value of `S`, including the one the designer never intended to use (`2'b11`), and values containing X or Z in simulation. The synthesized circuit (Figure 5.11) is two comparators and two 2-to-1 multiplexers in series.

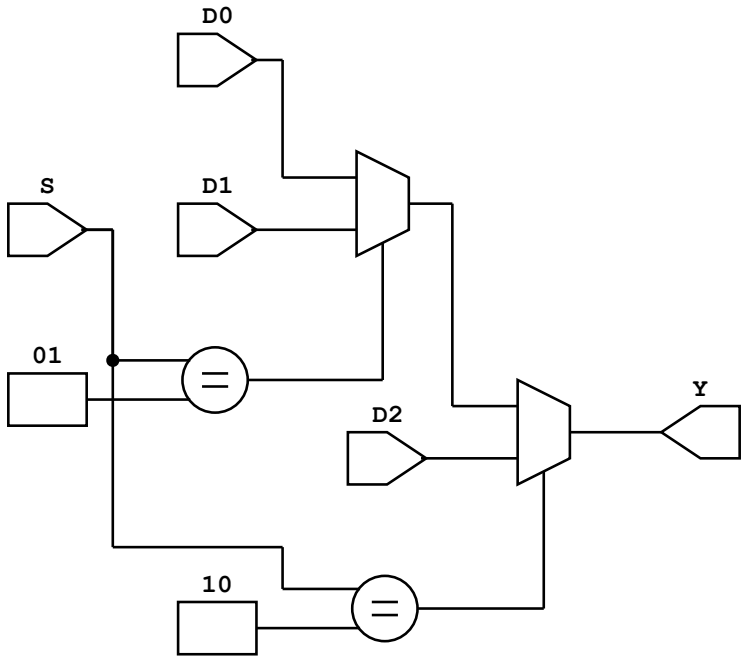


Figure 5.11: The `mux3` cell as drawn by Yosys from the HIPV file: `S == 2'b10` selects `D2`, otherwise `S == 2'b01` selects `D1`, otherwise `D0`.

5.12 The decoder

A **decoder** turns an encoded value into a set of separate control signals, and it is everywhere in a processor: the instruction decoder turns an opcode into control signals, and the bus decoder turns an address into "which device is being accessed". The second one is a clean example. The HIPV crossbar has five slaves (instruction memory, data memory, UART, timer and GPIO), each mapped to a region of the address space, and the decoder produces a **one-hot** select, one bit per slave, plus an error flag for an address that belongs to no slave. Here it is with the address map written out:

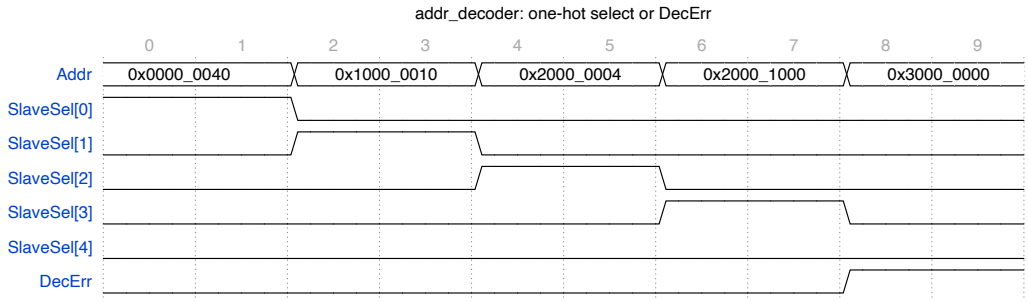


Figure 5.12: An address decoder: one `SlaveSel` bit goes high for each mapped region, and `DecErr` goes high for an unmapped address.

```
module addr_decoder (
    input  logic [31:0] Addr,
    output logic [4:0]  SlaveSel,
    output logic        DecErr
);
  always_comb begin
    SlaveSel = '0;                                      // default: nothing selected
    if (Addr[31:16] == 16'h0000)  SlaveSel[0] = 1'b1;   // instruction memory, 64 KiB
    if (Addr[31:16] == 16'h1000)  SlaveSel[1] = 1'b1;   // data memory, 64 KiB
    if (Addr[31:12] == 20'h20000) SlaveSel[2] = 1'b1;   // UART, 4 KiB
    if (Addr[31:12] == 20'h20001) SlaveSel[3] = 1'b1;   // timer, 4 KiB
    if (Addr[31:12] == 20'h20002) SlaveSel[4] = 1'b1;   // GPIO, 4 KiB
  end

  assign DecErr = ~|SlaveSel;
endmodule
```

Read the waveform first (Figure 5.12). Five addresses arrive one after another, two columns each. The first, `0x0000_0040`, lies in the instruction-memory region, so `SlaveSel[0]` goes high. The next addresses hit data memory, the UART and the timer in turn, and each raises exactly one select. The last, `0x3000_0000`, matches no region: every select stays low, so `DecErr` rises. Only one select is ever high, because the regions do not overlap.

Now the code. The first statement in the block is the important one: `SlaveSel = '0` assigns the default, *every bit low*, before anything else. Each `if` then overrides one bit. Because every bit of `SlaveSel` has a value on every path, whatever the address, the block cannot build a latch, and none of the `if`s needs an `else`. This **default first, then override** pattern is the safest way to write `always_comb`, and you will see it throughout HIPV. The last line

uses the reduction operator from the previous chapter: `~|SlaveSel` is true when no select bit is set.

(The real `obi_decoder` in the HIPV repository is the same block, with the base addresses and sizes taken from a shared package, so that adding a slave changes one file.)

5.13 Avoiding latches

The decoder showed the right way. Here is the wrong way:

```
always_comb begin
  if (sel) y = in1;
end
```

When `sel` is 0, no branch assigns `y`. In a spreadsheet an unhandled case would show a blank cell, but in hardware an output cannot go blank. If no branch assigns it, the only way to keep it at *some* value is to remember the previous one, so the tool builds a **latch**, the element from earlier in this chapter (Figure 5.13). It adds a storage element you did not intend, it complicates timing analysis, and it is the most common bug in `always_comb`. Tools warn about it, which is the reason to use `always_comb` instead of the old `always @(*)`.



Figure 5.13: What synthesis builds for an incomplete `if`: a latch whose enable is `sel`, so `y` holds its old value whenever `sel` is 0.

There are three ways to follow the rule. Give every output a **default** at the top of the block, as the decoder does. Or write an `else` on every `if`. Or give every `case` a `default` branch, as in `mux3`. Sequential blocks are different: inside `always_ff`, a missing assignment simply means “keep the value”, which is what a flip-flop does anyway.

5.14 Blocking vs. non-blocking assignment

We have used two assignment operators in an `always` block, and the pairing has been the same every time: `=` in `always_comb`, and `<=` in `always_ff`. Now we can see why.

A **blocking** assignment, `=`, takes effect immediately: the next statement in the block sees the new value. A **non-blocking** assignment, `<=`, only *schedules* the update, which lands at the end of the current time step: the next statement still sees the old value. The difference only shows when there is *interdependency*, when a later statement reads something an earlier one wrote.

5.14.1 In combinational logic

Take three inputs and the function `s = c ^ (a ^ b)`, written with an intermediate signal, once with each operator:

```
module xor_gate (
  input  logic a, b, c,
  output logic s1, s2
);
  logic n1, n2;

  // case 1: blocking
  always_comb begin
    n1 = a ^ b;
    s1 = c ^ n1;
  end

  // case 2: non-blocking
  always_comb begin
    n2 <= a ^ b;
    s2 <= c ^ n2;
  end
endmodule
```

(Some tools refuse non-blocking assignments inside `always_comb` altogether; the example is written this way only to show what would happen.)

Figure 5.14 shows the result for the same inputs. In case 1, `n1` updates immediately, and the next line uses the new `n1`, so `s1` is always `c ^ n1`. In case 2, `n2` is only *scheduled* to change, so `s2 <= c ^ n2` reads the **previous** `n2`. Compare the last two rows: `n` changes at the start of

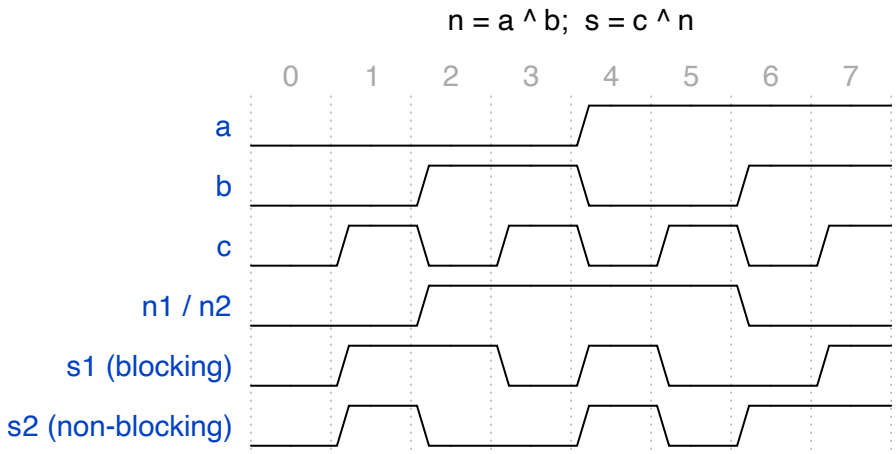


Figure 5.14: The two XOR descriptions: `s1` always matches `c ^ n1`, while `s2` is computed from the previous value of `n2`.

columns 2 and 6, and at exactly those two points `s2` differs from `s1`, because it still uses the old `n`. Case 1 models a combinational circuit; case 2 does not, because its output depends on a stale value. Synthesis tools often ignore the distinction in combinational blocks and build the right gates anyway, which makes it worse: the simulation shows one behavior and the hardware has another, a *simulation/synthesis mismatch*, the kind of bug that is hardest to find.

5.14.2 In sequential logic

The same experiment on two registers in a chain:

```
module chain (
  input  logic clk, a,
  output logic f1, f2
);
  logic b1, b2;

  // case 1: blocking
  always_ff @(posedge clk) begin
    b1 = a;
    f1 = b1;
  end
```

```
  // case 2: non-blocking
  always_ff @(posedge clk) begin
    b2 <= a;
    f2 <= b2;
  end
endmodule
```

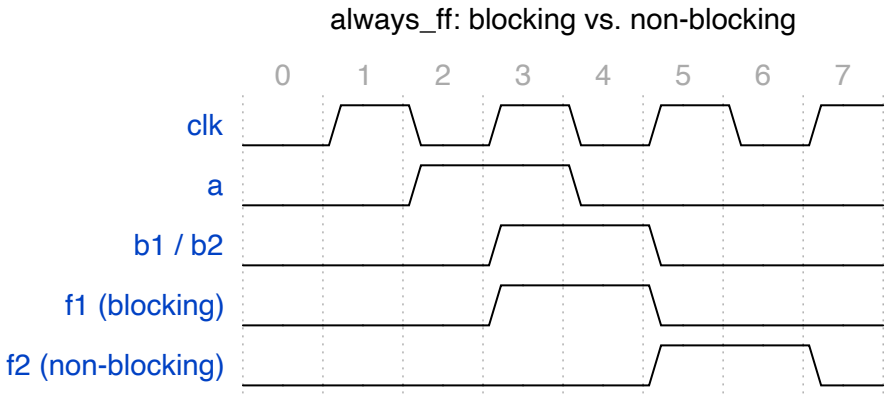


Figure 5.15: Two registers in a chain: with blocking assignment `f1` follows `b1` in the same cycle, with non-blocking assignment `f2` lags by one more clock.

In Figure 5.15 the clock is drawn with two columns per period, so it rises at the start of columns 1, 3, 5 and 7. The input `a` is high during columns 2 and 3, so the edge at the start of column 3 is the one that sees it. In case 1, `b1 = a` happens first, and `f1 = b1` then reads the *new* `b1`, so `f1` gets the value of `a` in the same cycle: both change at the edge starting column 3, and the second register adds nothing. In case 2, `b2 <= a` and `f2 <= b2` are both evaluated before either update lands, so `f2` takes the value `b2` had *before* this edge. `b2` rises at column 3, but `f2` only at the next edge, at column 5, and it falls one edge after `b2` does, at column 7. That is a two-stage chain, which is what two registers in a row should be. This is the behavior every pipeline relies on, and it is why the swap `a <= b; b <= a;` works with `<=` and silently fails with `=`.

5.14.3 The rule

	Blocking =	Non-blocking <=
Update takes effect	immediately	at the end of the time step
The next statement reads	the new value	the old value

	Blocking =	Non-blocking <=
Use in	`always_comb`	`always_ff`

= for combinational logic, <= for clocked logic, every time. Do not mix the two in one block.

5.15 Structural design: module instances

Behavioral modeling describes what a block *does*. **Structural** modeling describes a block by how it is *composed* of simpler modules. Each copy of a smaller module is an **instance**, and it has a unique name. Here is the smallest useful example, a 4:1 multiplexer assembled from three copies of a 2:1 multiplexer:

```
module mux2to1 (
    input  logic a, b, sel,
    output logic y
);
  assign y = sel ? b : a;
endmodule

module mux4to1_struct (
    input  logic [3:0] d,
    input  logic [1:0] sel,
    output logic       y
);
  logic y0, y1;                 // wires between the instances

  mux2to1 mux_low   (.a(d[0]), .b(d[1]), .sel(sel[0]), .y(y0));
  mux2to1 mux_high  (.a(d[2]), .b(d[3]), .sel(sel[0]), .y(y1));
  mux2to1 mux_final (.a(y0),   .b(y1),   .sel(sel[1]), .y(y));
endmodule
```

An instantiation has three parts. First comes the **module name** (`mux2to1`), then the **instance name** you choose (`mux_final`), then in parentheses the **port connections**. The notation `.a(y0)` reads “the module’s port `a` is connected to my signal `y0`”. These connections by name make the code easy to read and safe to reorder. There is also a positional form, `mux2to1 mux_final (y0, y1, sel[1], y);`, which is shorter and fails silently if someone later reorders the ports, so we avoid it. Figure 5.16 shows the result: the tool leaves the three instances as boxes, exactly as the hierarchy appears in the code.

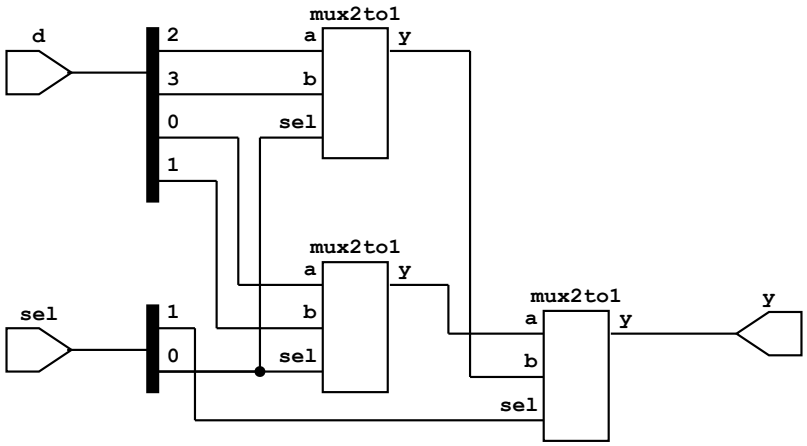


Figure 5.16: The structurally built `mux4to1_struct`: two instances select within a pair, and a third selects between the two results.

5.15.1 Parameters

A **parameter** makes a module configurable, which is how one description serves every width. We have been using this all along: `flopenr` declares `parameter WIDTH = 8`, and each instance picks its own. The override goes between the module name and the instance name, after a `#`:

```
flopenr #(32) pcreg (...);          // a 32-bit instance, by position
flopenr #(.WIDTH(32)) pcreg (...);  // the same, by name
```

The by-name form is safer when a module has several parameters. A `localparam` is a constant inside a module that, unlike a `parameter`, cannot be overridden from outside, which makes it the right choice for derived constants.

5.15.2 A real example: the program counter of HIPV

HIPV's fetch unit is almost entirely structural. Its program-counter loop is built from three of the leaf cells we wrote above. The code is the actual instantiation, reduced to this loop:

```
module pc_unit (
    input  logic        clk, reset,
    input  logic        PCEnF,
```

```
    input  logic        PCSrcE,
    input  logic [31:0] PCTargetE,
    output logic [31:0] PCF
);
  logic [31:0] PCNextF, PCPlus4F;

  // next PC selection: sequential or redirect from Execute
  mux2 #(32) pcmux (
      .D0(PCPlus4F), .D1(PCTargetE), .S(PCSrcE), .Y(PCNextF));

  // PC register
  flopenr #(32) pcreg (
      .clk(clk), .reset(reset), .en(PCEnF), .D(PCNextF), .Q(PCF));

  // sequential PC
  adder #(32) pcadd4 (
      .A(PCF), .B(32'd4), .Y(PCPlus4F));
endmodule
```

Everything inside the parentheses is a wire name, and nothing here describes any behavior: it only says which cell goes where. The register holds the PC, the adder computes `PC + 4`, and the multiplexer chooses between that and a branch target (Figure 5.17). Note how the signals form a loop through the register, which is the only place where the loop is broken by a clock.

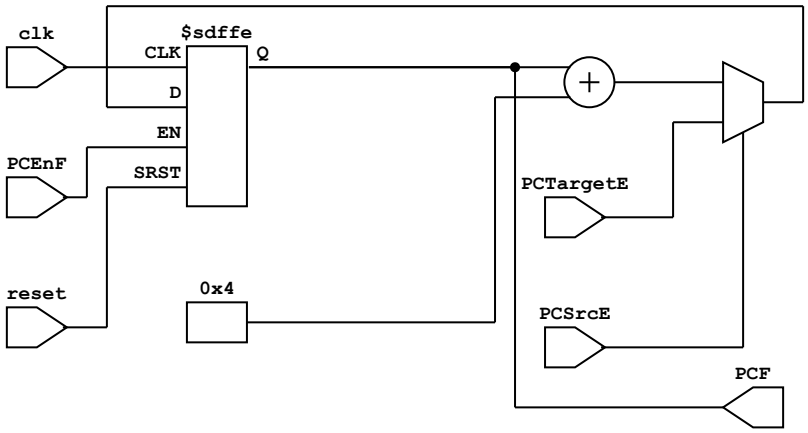


Figure 5.17: The PC loop of HIPV, drawn by Yosys with the three cells expanded into their hardware: the `$sdffe` register, the `+4` adder, and the next-PC multiplexer.

5.16 Testbenches

A **testbench** is a module that tests another module, the *device under test* (DUT). It applies inputs to the DUT and, ideally, checks that the outputs are right. Its inputs and the expected outputs are the *test vectors*. A testbench has no ports of its own: it sits at the top of the simulation, so everything it needs is declared inside as plain `logic` signals and wired to the DUT by name, like any other instance.

```
module tb_mux4to1;
  logic [3:0] d;
  logic [1:0] sel;
  logic       y;

  mux4to1_struct dut (.d(d), .sel(sel), .y(y));

  initial begin
    d = 4'b1010;
    for (int i = 0; i < 4; i++) begin
      sel = i[1:0];
      #10;
      $display("d=%b sel=%b y=%b", d, sel, y);
      if (y !== d[sel]) $error("sel=%b: expected %b, got %b", sel, d[sel], y);
    end
    $finish;
  end
endmodule
```

Everything happens in the `initial` block, which starts at time zero and runs once. It holds `d` fixed and sweeps `sel` through all four values. After each change it waits `#10` time units, `$display` prints the inputs and the output, and the `if` checks `y` against the bit of `d` that `sel` selects. The `#10` delay is not needed by the circuit, since a combinational output settles almost instantly. It gives each test case its own slice of simulation time, so the printed lines and the waveform stay readable. The comparison uses `!==`, which also catches an `X` on the output where `!=` would stay silent. `$finish` ends the simulation.

These statements (`initial`, `#10`, `$display`, `$finish`) exist only in simulation. A synthesizer has no hardware for “wait 10 time units”, which is the line between the code you simulate and the code you synthesize.

5.17 Testbench for a sequential design

A registered design only changes on a clock edge, so the testbench has to supply that clock; nothing else in the simulation will. Here it is for the counter of this chapter:

```
module tb_counter;
  logic       clk = 0;
  logic       reset = 1;
  logic [3:0] count;

  counter4 dut (.clk(clk), .reset(reset), .count(count));

  always #5 clk = ~clk;              // clock generator: period of 10 units

  initial begin
    #12 reset = 0;                   // release reset between clock edges
    #100;
    $display("count=%0d", count);
    $finish;
  end
endmodule
```

The separate `always #5 clk = ~clk;` has no sensitivity list and no guard, so it keeps re-executing for the whole simulation and flips `clk` every 5 time units. That gives a free-running clock with a period of 10. It runs concurrently with the `initial` block below it: every `initial` and `always` block in a module starts at time zero and proceeds independently, which is why the clock generator lives in its own block.

The stimulus holds `reset` high for the first clock edge (at time 5), then releases it at time 12, between edges, so that no input changes exactly when the clock rises, the same convention as in our waveforms. After the remaining ten rising edges the counter holds 10, which `$display` prints. The key difference from the combinational testbench is the waiting. A combinational output is valid almost immediately after its inputs change, but a register updates only once per clock edge, so a sequential testbench must always budget enough simulated time, or wait for a specific number of `@(posedge clk)` events, before checking the outputs.

`$finish` matters even more here. The clock generator never terminates on its own, so without an explicit `$finish` the simulation would run forever.

5.18 Running a simulation

Any SystemVerilog simulator can run these testbenches. With the open-source Icarus Verilog, you compile the design together with its testbench and run the result:

```
iverilog -g2012 -o sim_counter counter4.sv tb_counter.sv
vvp sim_counter
```

The `-g2012` option selects the SystemVerilog language level. To look at waveforms instead of printed lines, add two statements at the start of the `initial` block, and open the resulting file in GTKWave:

```
$dumpfile("tb_counter.vcd");
$dumpvars(0, tb_counter);
```

The waveforms you get are the same kind as the ones drawn in this chapter, and reading them is the same skill. Checking results by eye does not scale beyond a handful of signals, though. A testbench that compares outputs against expected values, as ours does with `$error`, can be run again after every change and report a clear pass or fail. Later in the course we move this idea to a Python framework, cocotb, which makes writing such checks much easier.

5.19 From design to the pipeline

Everything introduced here is the working vocabulary of the rest of the course. `always_ff` with non-blocking assignment and a synchronous reset is how every pipeline register is built, through `flopenr` and its siblings. `always_comb` with a default at the top is how the ALU and the decoders avoid latch inference. Instances and parameters are how those cells are assembled into stages, and stages into a processor. The pipeline chapters introduce no new syntax: they put these constructs to work on a larger example.

6 Five-Stage Pipeline — Structure

Five-stage pipeline

6.1 Goal for this week

- Translate the well-known pipeline theory (IF/ID/EX/MEM/WB) into SystemVerilog
- Apply SystemVerilog syntax immediately on a real, motivating example
- Treat memory (for now) as an interface — the implementation follows next week

We implement pipeline registers as separate modules for two reasons. First, pedagogically: each such register is a direct, visible example of a D flip-flop (DFF) in the context of a real processor, which was an explicit goal of the second part of the course. Second, practically: separate modules with a fixed interface (see `pipeline_pkg.sv`) allow us to substitute an individual stage with a reference ("golden") implementation in the jigsaw testing methodology — discussed in more detail in the verification chapter. This five-stage IF/ID/EX/MEM/WB structure follows the classic pipelined datapath described by Patterson and Hennessy (2017).

6.2 Modules for this week

- `if_stage`, `if_id_reg`
- `id_stage` (`regfile`, `imm_gen`, `decoder`)
- `id_ex_reg`
- `ex_stage` (`alu`)
- (`mem_stage`, `wb_stage` — interface only for now, implementation next week)

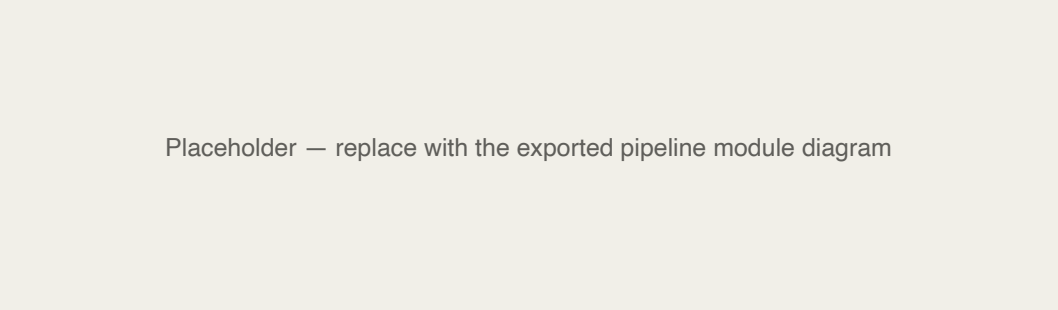


Figure 6.1: Modular pipeline structure

6.3 Example: `pipeline_pkg.sv`

```
package pipeline_pkg;

  typedef enum logic [3:0] {
    ALU_ADD, ALU_SUB, ALU_AND, ALU_OR, ALU_XOR,
    ALU_SLL, ALU_SRL, ALU_SRA, ALU_SLT, ALU_SLTU
  } alu_ctrl_t;

  typedef struct packed {
    logic mem_read;
    logic mem_write;
    logic [1:0] mem_size;
    logic mem_unsigned;
  } mem_ctrl_t;

  typedef struct packed {
    logic reg_write;
    logic [1:0] wb_sel;
  } wb_ctrl_t;

endpackage
```

6.3.1 Why a struct instead of individual signals

`typedef struct packed` is standard synthesizable SystemVerilog (IEEE 2017), so this pattern is portable across the tools you will use this semester.

An alternative to the approach above would be for each control signal (`mem_read`, `mem_write`, `reg_write`, …) to travel through the pipeline as a separate wire and a separate field in every pipeline register. This quickly becomes unwieldy as the number of control signals grows — every new signal requires a port change on every intermediate register. Using a `typedef struct packed` instead, we define a single type that a pipeline register carries as a whole; adding a new field to the struct does not require changing the register's ports.

6.4 Next step

The next chapter covers the hazards that arise from this pipeline structure: data hazards (resolved via forwarding or stalling) and control hazards caused by resolving branches in the EX stage, which requires flushing two instructions on a misprediction.

7 Five-Stage Pipeline — Hazards

Pipeline hazards

(placeholder — fill in with week 4 content: forwarding, stall, branch flush)

8 Memory — Implementation in SystemVerilog

Memory

(placeholder — fill in with week 5 content)

9 Verification with CocoTB

Verification with CocoTB

(placeholder — fill in with week 6-7 content)

References

IEEE. 2017. *IEEE Standard for SystemVerilog–Unified Hardware Design, Specification, and Verification Language*.

Patterson, David A., and John L. Hennessy. 2017. *Computer Organization and Design RISC-v Edition: The Hardware Software Interface*. Morgan Kaufmann.